VOCAL HEALTH · TECHNIQUE · INSPIRATION

VOICE



VOICE, SINGING AND SCIENCE

Speech-language pathology
applied to the singing
and speaking voice

SCIENCE TO UNDERSTAND.
ART TO MOVE.

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Voice, Singing and Science

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Voice, Singing *and Science*

Vocal health, vocal technique and inspiration: speech-language pathology applied to the singing and speaking voice

SPEECH-LANGUAGE PATHOLOGY APPLIED TO THE VOICE

Voice, Singing and Science
Vocal health, vocal technique and inspiration: speech-language pathology applied to the singing and speaking voice

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This book is intended for education and does not replace an assessment by health professionals.

“

Singers are their own instrument.

CHAPTER 9 · BREATHING AND SUPPORT

Contents

About This Book 08

About the Author 10

PART I Vocal Health 11

01 Speech-Language and Otolaryngological Assessment 12

02 Vocal Warm-Up and Cool-Down 17

03 Posture, Support, and Body Awareness 25

04 Vocal Hygiene and Voice Education 31

05 What Protects and What Threatens the Voice 36

PART II Foundations of Singing 46

06 The Vocabulary of Singing 47

07 Voice Classification 56

08 Singing Styles 60

09 Breathing and Support 69

10 The Physiology of Phonation 90

11 Exploring Your Own Voice 105

PART III Vocal Ornaments 112

12 Fry 113

13 Twang 116

14 Yodel 119

15 Vibrato 123

16 Melismas 126

17 Cry 128

18 Appoggiatura 130

PART IV Advanced Techniques 132

19 The Vocal Modes 133

20 Sound Colors and Effects 141

21 Belting 144

References 151

Exercise Audio 154

INTRODUCTION

About This Book

The voice is at once the most intimate instrument we have and the one we ask the most of — to sing, to teach, to sell, to lead, to comfort.

This book was written for anyone who wants to understand how the voice works and learn to use it with technique, expression and confidence. It brings together and updates the teaching material developed by Dr. Brízia Lacerda through her clinical practice and her work as a singing teacher, drawing on research and on the scientific and vocal-pedagogy literature. Sources are cited discreetly throughout the chapters, with the full list at the end.

Who this book is for

Anyone learning to sing

Breathing, support, registers, voice classification and exercises to train safely.

Voice teachers

Terminology, physiology, styles and ornaments, organized for reference and for the studio.

Anyone who wants to communicate well

Care for the speaking voice, hydration, vocal hygiene and guidance for public speakers.

Speech-language pathologists

Assessing the singer, risk factors, harmful habits and the physiological basis of phonation.

How the book is organized

Part I · Vocal Health covers assessment, warm-up and cool-down, posture, vocal hygiene and the factors that protect or threaten the voice. **Part II · Foundations of Singing** presents terminology, voice classification, singing styles, breathing and the physiology of phonation. **Part III · Vocal Ornaments** explores fry, twang, yodel, vibrato, melismas, cry and the appoggiatura. **Part IV · Advanced Techniques** introduces the vocal modes, sound colors, effects and belting.

Exercises are marked with a headphone symbol: every one of them comes with supporting audio, which you can download using the QR code and website on page 154.

> PLEASE NOTE
>
> This book is intended for education and does not replace an evaluation by a physician, a speech-language pathologist or a voice teacher. Symptoms such as persistent hoarseness, pain or loss of voice call for professional assessment.



THE AUTHOR

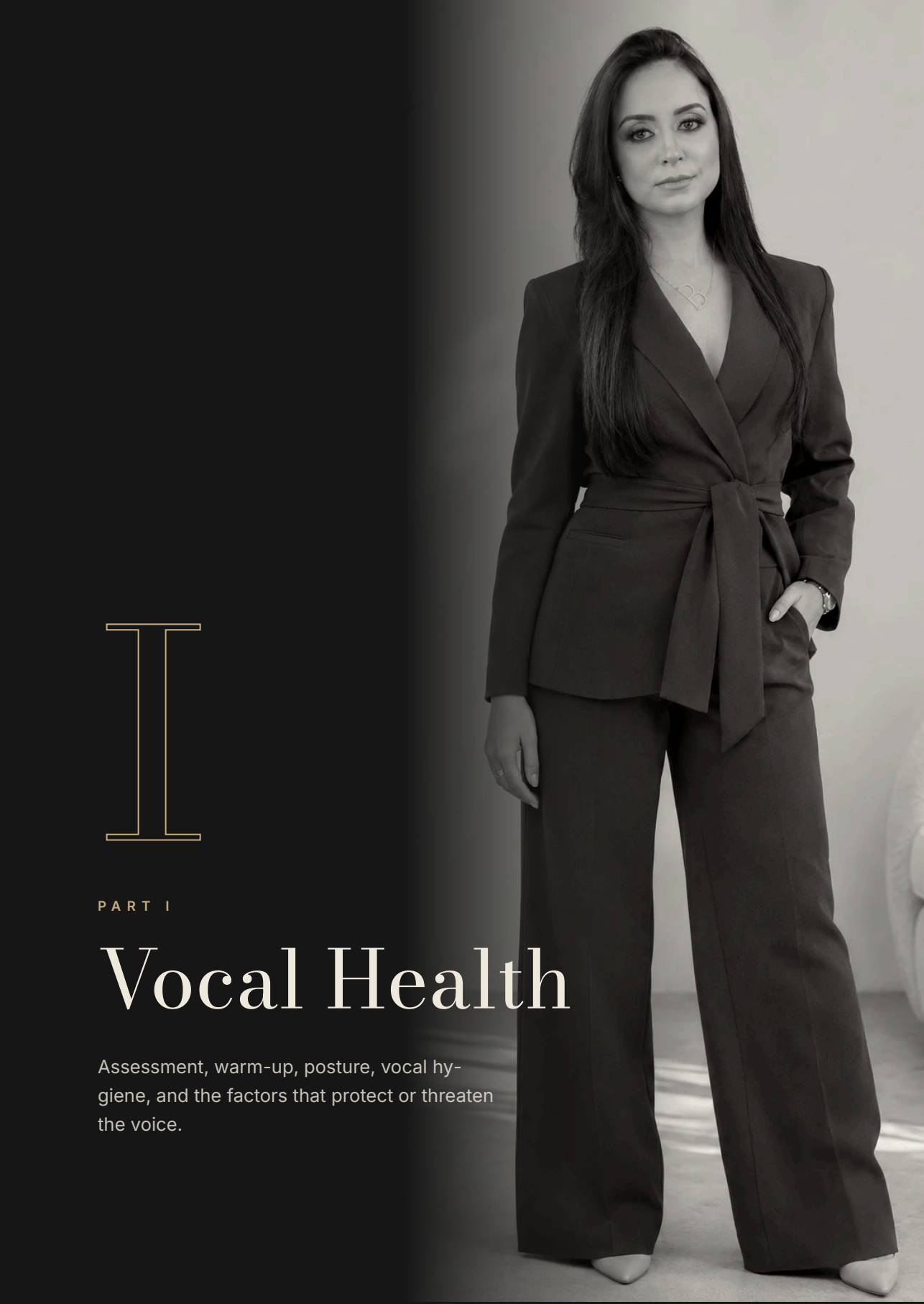
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A speech-language pathologist trained at PUC Minas (Pontifical Catholic University of Minas Gerais, Brazil), with postgraduate training and a specialization in voice. She works as a singer, vocal coach, researcher, writer and speaker, and is the author of a published scientific article.

As head of Faculdade da Voz, she brings together voice science, singing, teaching and mentorship for voice teachers, speech-language pathologists, singers and professionals who want to deepen their knowledge and practice of the voice.

A mother and a wife, she brings to voice teaching the same care with which she sees life itself: the voice as an expression of the whole person — body, emotion and purpose.

Brízia Lacerda



I

PART I

Vocal Health

Assessment, warm-up, posture, vocal hygiene, and the factors that protect or threaten the voice.

01

CHAPTER 1

Speech-Language and Otolaryngological Assessment

Of all voice professionals, singers place the greatest demands on their voices. Even the slightest vocal problem is enough to make a singer tense and anxious: however minor it may be, it will have a major impact on their singing, potentially leading to vocal strain and hoarseness. For this reason, singers tend to be the group of professionals most focused on prevention and voice care.

At the first sign of a symptom, a singer should seek professional help. Sometimes the first stop is a speech-language pathologist; at other times it is a general physician or an otolaryngologist (ENT). The most cautious singers see both professionals regularly, even when nothing is wrong, in order to prevent injury and improve their vocal conditioning.

Assessment of the singing voice essentially comprises the singer's case history, which gathers important, relevant information for the diagnostic workup and subsequent treatment, together with a clinical examination by an otolaryngologist to evaluate the larynx and its respiratory, sphincteric, and phonatory functions.

The otolaryngology consultation and the laryngeal examination are inseparable from a detailed case history. The history reveals factors that can alter the voice on their own, that may also aggravate other conditions and slow the patient's recovery, and that may call for specific therapeutic intervention. Likewise, it is necessary to understand the patient's overall clinical picture and the possible physical and psychological causes that can affect the larynx and the voice.

It is essential to analyze factors specific to the vocal apparatus alongside general health conditions, such as: diabetes mellitus; high blood pressure; reduced pulmonary ventilatory capacity; heart failure; metabolic, endocrine, allergic, immunological, autoimmune, neurological, infectious, inflammatory, and digestive disorders, among others; benign or malignant tumors; the use or abuse of substances such as alcohol, caffeine, tobacco, tranquilizers, marijuana, cocaine, and others; the ongoing or prolonged use of medications such as corticosteroids, anticoagulants, antiplatelet agents, and chemotherapy drugs; and states such as anxiety, depression, fatigue, stress, burnout, insomnia, pregnancy, and premenstrual syndrome.

Both the physician and the speech-language pathologist should consider the singer's vocal characteristics and complaints separately as they relate to speaking and to singing.

Although a single organ and essentially the same group of muscles are responsible for both functions, several fundamental aspects of voice production differ greatly between the speaking voice and the singing voice. In many cases, singers need a therapist's or physician's help to pinpoint their complaint, distinguishing the characteristics of the speaking voice from those of the singing voice. It is common for someone to

see a professional with a complaint specific to the singing voice, such as difficulty with a register transition in a scale, or, conversely, to have no awareness at all of how they use their speaking voice.

FOR THE SPEECH-LANGUAGE PATHOLOGIST WHO WORKS WITH SINGERS

Assessing a singer goes beyond assessing the speaking voice. It includes the singer's vocal history (style, repertoire, hours spent in rehearsals and performances, use of microphones and stage monitors), the complaint as it appears in real performance settings, and observation of the singing voice across different registers. Each of the factors listed below can be explored during the consultation and monitored over time.

Key factors in assessing the singing voice

- **Hearing disorders.** Any hearing impairment can compromise the singer's entire feedback loop, the ability to hear what they are singing. Not only the voice but also hearing should be monitored, through specific tests ordered by the physician or the speech-language pathologist.
- **Types of voice disorders.**
- **Anatomical variations of the head and neck.** These may lead to changes in resonance and posture, and to compensatory movements during phonation.
- **Neck sensitivity.** This can cause vocal tension, poor posture, and a lack of harmony between body and voice.
- **Posture.** During spontaneous conversation, check the tilt of the head, tension in the neck muscles, the position of the chin relative to the chest, and the slope of the shoulders. Observe the curvature of the spine while the singer walks and sits, and determine whether the

posture they adopt is suitable for phonation. It is also important to see whether the postural traits observed during speech change or remain the same during singing. Many people raise their chin when singing higher notes, in the mistaken and harmful belief that this helps the larynx rise. In fact, a larynx that rises freely, without tension and without the involvement of the extrinsic muscles, is what would genuinely help in producing high notes.

“

No singing exercise should ever be uncomfortable.

CHAPTER 2 · VOCAL WARM-UP AND COOL-DOWN

02

CHAPTER 2

Vocal Warm-Up and Cool-Down

Exercising your voice safely

CAUTION

— **No singing exercise should ever be uncomfortable.** If you feel any tension, pain, burning, or discomfort, stop immediately. Always practice carefully and attentively so that you notice any of these signals while you work. Do not push through: you could injure yourself. Stop, then start again to recover the correct posture or position for the exercise.

— **Avoid constriction.** Notice whether you are jutting or clenching your jaw, or whether there is any constriction in your lips, throat, neck, and/or shoulders. Do not let a lack of breath support lock up your muscles. Strengthen your support and counter-support, and relax your upper body.

Vocal warm-up

A vocal warm-up prepares the body to perform its functions properly and helps prevent injury by reducing friction between the vocal folds and tears in the muscle fibers and connective tissue. Warm-up exercises increase blood flow, which supports greater intensity, vocal projection, clearer articulation, and richer production of harmonics.

Suggested exercises:

EXERCISE · NASAL SOUNDS /M/ AND /N/

These can be combined with tongue movements and chewing.

- "Ummmm" (sliding from high to low and from low to high).
- "Ummmm" combined with chewing, both voiced and unvoiced.
- Continuous production of "m" (mouth closed), "n," or "ny" (as in "canyon"), sustained, modulated, or sung on scales.
- M...BRRRR...M...BRRRR...M...BRRRR / M...DRRRR...M...DRRRR (a nasal sound combined with a trill).
- Humming "mmm..." in a yawn position with the mouth closed, to create internal space and increase resonance.
- "Mini-mini-mini-mini...aaaaaaaa" and "mah-nah-nyah, meh-neh-nyeh..." on a chanted voice: fine tuning through small, repeated emissions to train short phonatory units and a stable onset.
- A nasal sound produced together with tongue clicks.
- The siren technique: with the tongue in the position for the nasal "ny," keep the mouth cavity closed off and glide through a wide range of pitches.

Benefits: softens the voice, reduces laryngopharyngeal resonance, increases maximum phonation time, and improves projection.

Audio for this exercise available for download · see page 154

EXERCISE · TONGUE AND LIP TRILLS

Keep the tip of the tongue vibrating on “TRRR,” prolonging the sound and ending on a vowel:

TRRR... A · TRRR... E · TRRR... I · TRRR... O · TRRR... U

Repeat the same process with the lips vibrating on “BRRR,” prolonging the sound and ending on a vowel:

BRRR... A · BRRR... E · BRRR... I · BRRR... O · BRRR... U

Benefits: mobilizes the mucosa of the vocal folds, balances the coordination of breathing, phonation, and articulation, and reduces phonatory effort.

Audio for this exercise available for download · see page 154

EXERCISE · VERY-HIGH-PITCH TECHNIQUE

— Counting numbers.

— Vowels.

Benefits: relaxes the thyroarytenoid muscles, engages the cricothyroid muscles, and balances resonance in the modal register.

Audio for this exercise available for download · see page 154

EXERCISE · VOCALIZING ON VOWEL SEQUENCES

ee · ay · eh · ah · aw · oh · oo

eeeeeeeee · ayyyyy · ohhhhh · ooooo

Benefits: improves control of the glottal source, reduces constriction in the vocal tract, sharpens vowel definition, stabilizes vocal quality, heightens vocal awareness, and balances resonance.

Audio for this exercise available for download · see page 154

EXERCISE · ARTICULATION EXERCISES

For the lyrics of a song to be understood, words and speech sounds must be articulated precisely. Shaping those sounds requires agile, toned, and released muscles.

The lips, cheeks, and tongue all need to be trained so that each sound can be placed at its correct point of articulation. Nor should the teeth be forgotten: they need proper care, both in terms of hygiene, to avoid inflammation and discomfort, and in terms of a healthy bite that does not strain the temporomandibular joint. All of these factors contribute to comfortable articulation.

A cork can be used to intensify the exercise. Bite down on the cork just enough to hold it in place. With the cork in place, read the tongue twisters below: first speak only their vowel sounds, then say the full sentences.

These English tongue twisters were chosen as equivalents of the original Portuguese set, working the same consonant clusters (pr, br, tr, cr, dr, gr, fr, fl, pl, bl, tl, cl, gl).

a. The prestigious professor promptly prepared a profoundly practical presentation.
b. Pretty Priscilla prances proudly past the prickly prairie primroses.
c. Crisp crusts crackle and crunch as crafty Cressida crumbles crackers.
d. Brave Brian broke the brittle brown bread on the bright brick bridge.
e. Brenda's brothers bravely brushed the brindled bull's bristly brows.
f. Trevor trusted the trembling troubadours to trade true tales of treasure and trouble.

g. Drowsy Drew dreamed the dragon drank the dreary drizzle dripping down the drain.

h. Proud Pedro dropped three dripping stones from the prickly quarry and tripped on the driftwood.

i. Crazy crows crowd the crooked crypt, cracking crimson crab claws.

j. Green grapes grow in great green groves where grumpy gray groundhogs grin and grumble.

k. Fresh fried fish, fish fresh fried, fried fish fresh, fish fried fresh.

l. A flea and a fly in a flue were imprisoned, so what could they do? Said the fly, “Let us flee!” “Let us fly!” said the flea, so they flew through a flaw in the flue.

m. Plucky planes plunge and plane above the plateau, where plain planters plant plump plums.

n. Please place the plaque on the plump, pleated pillow, and plead for a placid plan.

o. Bleak black blizzards blow blue blankets over blushing, blond Blake.

p. An athlete crossed the Atlantic in search of the lost Atlantis she had glimpsed in an old atlas.

q. Clever Clara clearly declared that the cloudy clock was clicking clumsily.

r. Gleeful Gloria glimpsed glittering globes glowing in the glassy glen.

Benefits: reduces hypertonicity and increases volume, projection, vocal endurance, and articulatory precision.

Audio for this exercise available for download · see page 154

EXERCISE · DEEP-BREATHING EXERCISES

— Alternate between mouth and nose breathing; lengthen the spine while coordinating inhalation and exhalation.

— Move the neck muscles (side tilts, forward and backward tilts, and rotation).

Audio for this exercise available for download · see page 154

EXERCISE · PALATE SWEEP

Run the tip of your tongue along the roof of your mouth (the hard palate) back toward the soft palate, then circle the tongue around the vestibule, sweeping the “walls of the mouth,” while keeping your lips together.

Benefits: lowers the larynx and relaxes the extrinsic laryngeal muscles.

Audio for this exercise available for download · see page 154

EXERCISE · VOCALISES

These are vocal exercises sung with an instrument, which keeps the singer in tune, trains the ear, and works the vocal folds at the same time.

Benefits: extends vocal range, promotes the lengthening and shortening of the vocal folds, refines the voice, and heightens awareness of pitch.

Audio for this exercise available for download · see page 154

Special attention to the larynx

The larynx is one of the most important and delicate structures in our vocal system. It is extremely important that you learn to sense how it feels. To do so, use your own hand to move the larynx gently from side to side and up and down.

EXERCISE · EXERCISES FOR THE LARYNX

— **Silent yawn:** yawn without making a sound, keeping the tip of the tongue from pulling back as it naturally would.

— **Little gasp:** draw in air very gently, with just a hint of sound, as if mildly startled.

— **Vocal flex:** open and close the throat (moving the vocal folds) very gently, without breathing.

— **Steamboat:** blow out with puffed cheeks, making a very soft sound.

— **Grimace:** stick your tongue out of your mouth forcefully, then relax.

— **Fish lips:** move the orbicularis muscles of the lips in a relaxed way, silently overarticulating the vowels "OO-AH-EE, OO-AH-EE" (like a drawn-out "why, why").

— **Grimace with fish lips (chewing):** combine the last two movements, using the tongue and the lip muscles at the same time, while counting softly to 10.

Audio for this exercise available for download · see page 154

> *"All of these exercises should be performed with close attention to breathing and to the use of the diaphragm, controlling the exhaled air without forcing the throat."*

Vocal cool-down

A vocal cool-down readjusts the speech and articulation system from the singing voice back to the speaking voice, reducing phonatory effort and softening the voice.

- **Yawning technique:** yawning reduces hard glottal attacks and medial glottal compression, widens the vocal tract, lowers the larynx, improves source-filter tuning, and raises the soft palate.
- **Head rolls on the vowels /a/, /o/, and /u/:** dynamic relaxation and body-voice integration.
- **Nasal and/or trilled sounds on descending glides:** soften the voice and reduce laryngopharyngeal resonance.
- **Chanted voice:** reduces overall effort.
- **Fingertip massage of the face and neck:** reduces hypertonicity and lowers the fundamental frequency.
- **Spontaneous speech, such as recounting an experience,** to discriminate between phonatory settings (singing versus speaking): readjusts the speech organs for the speaking voice, stabilizes vocal quality, and balances resonance.

Audio for the cool-down exercises available for download · see page 154

03

CHAPTER 3

Posture, Support, and Body Awareness

One of the most common concerns in a singer's vocal work is the voice “going to the throat,” the place where vibration happens. More specifically, we are talking about the vocal folds, which vibrate to produce sound and should therefore be kept free of any tension.

Nevertheless, a point of support is needed to project the voice as far as one wishes (GAVA JÚNIOR; FERREIRA; ANDRADA E SILVA, 2010). As a rule, singers turn the throat into that point of support. This contradiction is why they end up with bulging throats and protruding veins. Our approach to support is the pressure that can be generated between the two diaphragms: the thoracic diaphragm and the pelvic diaphragm. The singer should apply slight abdominal pressure in the lower belly, roughly four finger-widths below the navel, followed by a slight contraction of the glutes (BJORKNER, 2008).

It is a bit like the subtle tummy tuck people do at the beach when someone pulls out a camera: a light drawing-in of the belly, without involving the stomach area, followed by a subtle squeeze of the glutes. This

should become part of the singer's daily life, a natural, constant habit. It is to this pelvic region that singers should direct their attention whenever they want stronger, more sustained projection (COELHO, 1994).

The pelvic girdle support, or "pelvic belt," as we like to call this bodily engagement, is also indispensable for stage speech. When an actor moves into shouting or whispering, maximum engagement of the pelvic belt is recommended. In this situation, negative pressure draws a large volume of air into the lungs. Actors very often raise their shoulders before shouting without pelvic support, placing that support in the throat instead; great care, observation, and practice are therefore recommended. When support is handled by the pelvic belt, the ribs take charge of drawing in and expelling the column of air (BJORKNER, 2008).

As air passes through the vocal folds, it sets them vibrating and produces the fundamental, or initial, sound, which is then amplified and shaped as it continues along its outward path. The paranasal sinuses and the sternum play a major role in amplifying the sound. Shaping is the work of the lips, the tongue, the soft palate, and the other structures of the mouth and nose.

EXERCISE · **PELVIC BELT EXERCISE**

- Draw the belly in (four finger-widths below the navel or, if you prefer, just above the pubic bone).
- Engage the glutes, squeezing and releasing the buttocks.
- Keep the knees soft and unlocked.
- While walking, squeeze the buttocks alternately.

Audio for this exercise available for download · see page 154

Body and interpretation

When we sing, we cannot always summon the emotion needed for the body to respond naturally; so we often fall back on forced, stereotyped gestures that convince no audience. Moreover, points of tension anywhere in the body are undesirable, because any physical imbalance can compromise vocal quality and performance.

> *"Hence the need to become aware of the body and to free it, so that there can be true 'surrender' in the interpretation [...]. If I am aware of my movement, my body will know how to behave and convey the intended message... When you truly believe in what you are singing, the emotion and feeling are transmitted from the inside out, so that the body speaks, whether still or in motion [...]"*
>
> MARSOLA; BAÊ, 2000

Stretching and loosening the body

EXERCISE · BODY EXERCISES

— **Body:** standing with your legs together and your arms at your sides, begin to inhale through your nose. By the end of the inhalation, your hands should be joined and your arms stretched overhead, with your heels raised as high as possible. Hold your breath for 3 seconds, then release the air gently through your mouth, as if blowing, while lowering your arms and heels.

— **Neck:** standing or seated, with your arms raised out to the sides at shoulder height and your hands on your chest, begin a slow inhalation through your nose while stretching your neck forward until your chin touches your chest. Hold the air for 3 seconds, then

return to the starting position, blowing the air out gently and lengthening your neck.

- **Head rolls:** gently turn your head to the right and to the left; then tilt it to each side, forward, and back, repeating each position four or five times and feeling the stretch in your neck muscles.
- **Face:** inhale and, with your mouth closed, humming on “m,” move your facial muscles slowly and broadly, as if chewing.
- Massage and activate the soles of your feet with the palms of your hands.
- Rotate your ankles.
- Loosen your knees: bend them slightly, place your palms on them, and circle them from side to side.
- Make circles with your hips, then a sideways figure eight.
- Bend your torso forward and let it hang, swaying gently from side to side.
- Raise your shoulders one at a time, letting each drop gently. Repeat with both together.
- Stretch your body up onto one foot while the other foot moves as if kneading the floor. Repeat on the other side.
- Circle your tongue inside your mouth 10 times in each direction. Move it quickly from side to side and, finally, lower your head, letting your tongue hang out as far as you can and holding that position for a few seconds. Raise your head without straining, gently drawing your tongue back in. Repeat a few times.

Audio for this exercise available for download · see page 154

EXERCISE · FOUR MOVEMENTS TO RELAX THE NECK

- **The “yes” movement:** up and down.
- **The “no” movement:** left to right.
- **The “maybe” movement:** ear toward shoulder, alternating sides.
- **Circles:** in both directions.

Audio for this exercise available for download · see page 154

“

Air is the fuel of voice production.

CHAPTER 4 · VOCAL HYGIENE AND VOICE EDUCATION



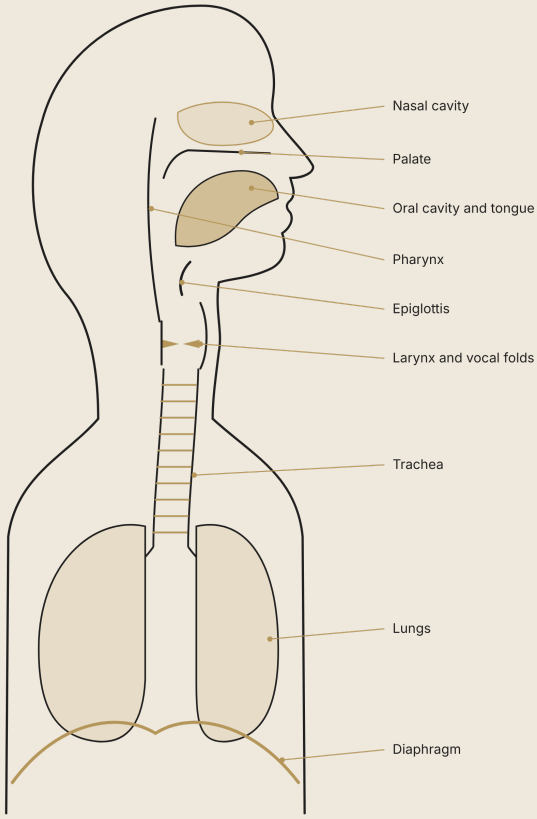
04

CHAPTER 4

Vocal Hygiene and Voice Education

Voice education includes explaining how phonation works and what vocal health means. Simple, accurate explanations of how sounds are produced, together with the basic rules of vocal hygiene and correct voice use, help raise the patient's awareness, enabling them to avoid episodes of dysphonia and to curb vocal abuse.

Vocal hygiene programs rest on the premise that vocal abuse contributes to the development of voice problems, and that eliminating or reducing these harmful behaviors therefore supports vocal rehabilitation and prevention. Numerous reports in the literature indicate that vocal hygiene programs are effective in preventing and eliminating vocal abuse.



The vocal apparatus: from the lungs to the lips, every structure involved in producing the voice.

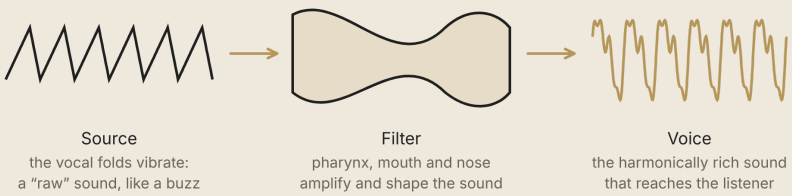
The explanation can conclude by showing how the different speech sounds are articulated in the mouth and why it matters that these movements be precise, so that speech is clear and intelligible and the message gets across.

How the voice is produced

Voice production begins in the brain, which sends impulses to the larynx, the organ that produces the voice. The larynx sits at the front of the neck, is shaped like a tube, and can be felt with the fingers. Inside this tube are the vocal folds, popularly known as the vocal cords. The term “vocal cords” is inaccurate, because they are not strings but two folds of tissue, lying parallel to the ground, that vibrate as air from the lungs passes between them.

You can feel these structures vibrate by placing your hand on your larynx while producing any sound, such as a prolonged “ah.” Producing voice, then, requires a command from the brain and air setting the vocal folds into vibration. Air is the fuel of voice production. Any imbalance in this mechanism, such as too much force or too much air, can lead to a voice problem. If too much air comes from the lungs, the voice will be breathy and the air will be audible in the sound; if, on the other hand, more muscular force is used than necessary, the sound will be squeezed into too little air and the voice will come out strained or hoarse.

SOURCE AND FILTER: HOW VOICE IS BORN



The source–filter model: the vocal folds generate the basic sound, and the vocal tract turns it into the voice we hear.

The sound generated in the larynx is not the voice we hear in the room: it resembles the buzz of an electric shaver. This raw sound passes through a natural loudspeaker formed by the pharynx, mouth, and nose, where it is amplified. These structures also modify the raw sound, mainly through movements of the tongue and lips, and in this way the vowels and consonants that form words are produced.

Vocal health can be defined as a person's ability to vary their voice in quality, pitch, loudness, and inflection according to the setting, the situation, and the context of communication.

Hydration

Mainly since the 1990s, adequate hydration has been recognized as a vocal hygiene measure and an excellent adjunct to voice therapy. Hydration treatment includes a range of measures, such as humidifying the environment, inhaling cool or warm vapor directly through the mouth and nose, breathing in fine water droplets, and taking mucolytic drugs (guaifenesin) to thin secretions.

Measures to control dehydrating conditions have also been suggested, such as avoiding very dry environments, smoking, alcohol, caffeine and other diuretics, antihistamines, nasal sprays and decongestants, and mouth breathing. For hydration, we generally recommend sipping water throughout the day (rather than drinking whole glasses at once), especially during sustained speaking. Cyclists' water bottles work well for this, since their spouts are very convenient to use and carry.

Another very useful strategy is to breathe in, through the nose, fine droplets of water from a piece of gauze or a thick paper tissue moistened with filtered or distilled water. The inhaled moisture travels directly to the larynx, and the droplets provide instant lubrication, helping to reduce the viscosity of the mucus. Naturally, your airways must be clear, free of allergic, inflammatory, or infectious conditions, before you

try hydration by inhaling water. Both drying conditions and overhydration should be avoided; overhydration can even worsen vocal conditions by depleting the body's mineral salts.

RESEARCH UPDATE · HYDRATION

A systematic review published in the *Journal of Voice* concluded that systemic hydration (drinking water regularly throughout the day) is the simplest and most cost-effective way to improve voice quality, and that it should be part of every vocal hygiene program. Steam inhalation improved acoustic measures such as *jitter*, *shimmer*, and the noise-to-harmonics ratio, and nebulization with isotonic saline lowered phonation threshold pressure. Mouth breathing, by contrast, which dries the surface of the vocal folds, worsened these same measures as well as perceived phonatory effort (ALVES et al., 2019).

FOR PUBLIC SPEAKERS

Teachers, lecturers, salespeople, customer-service staff, and communicators in general rely on their voices as working tools just as much as singers do. The principles in this chapter apply to you: sip water throughout the day, pause and vary your pace when you speak, avoid competing with background noise, breathe through your nose whenever possible, and warm up your voice before speaking for long periods. Clear communication depends less on speaking loudly and more on articulating well, projecting with support, and varying your intonation.

05

CHAPTER 5

What Protects and What Threatens the Voice

Many everyday factors affect the voice: smoking, alcohol, pollution, drugs, allergies, poor vocal habits, air-conditioning, poor diet, lack of rest, ill-suited clothing, high-strain sports, hormonal changes, and medications. This chapter examines each of them in turn.

> **A NOTE TO THE READER**
>
> The information in this chapter, especially regarding medications, is for educational purposes only and does not replace an evaluation by a physician or a speech-language pathologist. Do not start, stop, or change any medication without professional guidance.

Smoking

Smoke damages the entire respiratory system, the vocal folds above all, and can cause irritation, swelling, coughing, increased secretions, and infections. It also dries out the vocal folds, making it harder for them to vibrate. Smoking is considered one of the triggering factors for laryngeal and lung cancer, precisely because of the severe irritation it causes in these organs. Nonsmokers exposed to cigarette smoke can also develop problems, so not smoking indoors is a matter of respect for other people's health.

Alcohol

Drinking alcohol, especially distilled spirits, irritates the vocal apparatus much as cigarettes do, but its main effect is immunosuppression, that is, a weakening of the body's defense responses. Because the first drinks loosen the brain's cortical control, the drinker feels more uninhibited. Alcohol also mildly numbs the pharynx, and with sensation in this area reduced, a series of vocal abuses can go unnoticed. The consequences become evident only after the alcohol wears off.

Distilled spirits such as whiskey, vodka, cachaça (Brazilian sugarcane spirit), and brandy are worse for vocal health, irritating the tissues more intensely, especially when mixed with fermented drinks such as beer, champagne, and wine. There is a strong association between heavy consumption of distilled spirits and laryngeal cancer. The damage caused by alcohol and tobacco varies with the amount consumed, but other factors also influence the extent and severity of disease, including individual, constitutional, genetic, and family characteristics.

Finally, the habit some singers have of gargling with or drinking alcohol to “warm up the voice” is unwise. The surface effect of alcohol in the mouth and pharynx is a temporary numbness that genuinely dulls un-

pleasant sensations. As a result, the effort required to speak or sing goes unnoticed, and once the alcohol wears off, the consequences may include irritation of the vocal apparatus or a decline in voice quality.

Pollution and noisy environments

Pollution can cause acute or chronic changes in the voice and larynx. The most extreme cases involve fires, smoke, and chemical leaks, which can injure the entire respiratory tract. The use of dry ice and pyrotechnics at concerts and performances also irritates the vocal tract and can even injure the larynx; performers who work with theatrical smoke report more complaints and respiratory symptoms.

The link between pollution and voice disorders is not easy to prove, because it is complex and not necessarily direct: some pollutants reach the body directly when inhaled and deposited in the airways, while others act indirectly, through the bloodstream. Recent studies point to a higher incidence of airway cancers among people who work in places with airborne toxins and pollutants, such as insecticides.

Vocal and laryngeal symptoms include hoarseness, a feeling of irritation in the throat, coughing, difficulty breathing, and irritation of the tissues of the mouth, tongue, nose, and respiratory tract.

Exposure to noisy environments (noise pollution) can endanger not only hearing but also the voice. In a noisy setting, we reflexively raise our voices in an effort to communicate and be heard over the background noise: this is known as the Lombard effect.

Drugs

Marijuana is extremely harmful, irritating the mucosa not only through the smoke itself but also through toxins released by the burning paper in which the herb is rolled. In addition, pinching the joint between the

fingers and the teeth while smoking greatly raises the temperature of the smoke, damaging the tissues of the vocal tract. Besides a lowering of vocal pitch, users may show imprecise articulation and changes in speech rhythm and fluency.

Snorting powdered cocaine can directly injure the mucosa of the vocal tract through irritation and marked vasoconstriction. Perforations of the nasal septum and ulcerations of the vocal fold mucosa are common. Cocaine use can also alter perception and sensory control, thereby reducing control of the voice and leading to vocal abuse.

Injected cocaine causes muscular hypotonia, producing vocal fatigue and making it difficult to sustain adequate, efficient communication, particularly in professional voice use.

Allergies

People whose allergies affect the respiratory tract are more prone to voice problems, in direct proportion to the severity of the allergy. The respiratory mucosa tends to swell, which hinders the free vibration of the vocal folds, and constant secretions can irritate the larynx directly.

Nasal obstruction and persistent swelling caused by allergies can impair nasal resonance, making it harder to project the voice into the space. This difficulty can in turn lead to compensatory vocal strain, especially in children and voice professionals. People with allergies, particularly voice professionals, should therefore follow their doctors' instructions carefully to prevent flare-ups.

The effects of allergies are especially limiting for singers, who may struggle with high notes, experience breaks in the voice, and lose vocal agility.

An additional problem is the use of medications such as decongestants and antihistamines, whose side effects, including drying of the mucous membranes, are harmful to the voice. It is therefore essential to keep

allergies under control and, if possible, to identify the trigger. Environmental control is a powerful tool with no adverse side effects.

During an allergy flare-up, the speech-language pathologist's approach is to advise the patient to limit voice use to what is necessary while increasing hydration, so as to thin secretions and improve the conditions for mucosal vibration. When dryness is severe and the voice must be used professionally, topical hydration is recommended in addition to drinking fluids: steam inhalers used through the mouth and nose, or actively breathing in droplets of filtered water through the nose from soaked gauze held in the palm of the hand near the nostrils before heavy voice use.

Medications

The main medications to avoid because of their adverse effects on the voice:

- **Pain relievers:** avoid medications containing acetylsalicylic acid, as they can cause vocal fold hemorrhage. For example, many patients take one or two Aspirin tablets a day without medical advice, having heard about the benefits of thinning the blood. Acetylsalicylic acid is known to promote tissue bleeding and can lead to submucosal hemorrhages in the vocal folds, contact ulcers, or small frank bleeds; it is also a known cause of stomach irritation and ulcers.
- **Nasal sprays:** when overused, can cause dryness and swelling.
- **Cough suppressants:** highly irritating; can dry out the vocal tract.
- **Decongestants, antihistamines, and corticosteroids:** reduce respiratory tract secretions and cause dryness, as well as insomnia, irritability, stomach irritation, and tremors. The habit of receiving monthly injections, or an injection at every minor vocal change, can cause irreversible damage to the larynx, including atrophy of the vocal fold mucosa.

- **Antidiarrheals:** also reduce secretions and dry out the mucous membranes.
- **Diuretics:** reduce secretions, producing thick mucus and persistent throat clearing.
- **Vitamin C:** in high doses, can dry out the vocal tract.
- **Tranquilizers, sedatives, and sleep aids:** can interfere with the neural control of voice and speech production, affecting control of phonation and speech rhythm.

Air-conditioning

Tolerance of air-conditioning varies from person to person, but in general it is hard on the mucosa of the vocal folds: cooling the air also lowers its humidity, which dries out the vocal tract and leads to voice production with more tension and effort. If air-conditioning cannot be avoided, as in office buildings or in cars in large cities, the units should be properly adjusted and their filters cleaned regularly, and you should drink more room-temperature water.

Central heating and space heaters have the same drying effect. In these cases, small containers of water placed around the room can help raise the relative humidity, alongside good vocal hydration.

A healthy diet

Heavy, highly spiced foods slow digestion and restrict the free movement of the diaphragm, a muscle essential to breathing. Much of the body's energy is diverted to digestion, and vocal function suffers as a result. In addition, heavily spiced food promotes gastroesophageal reflux, especially with evening meals, which can irritate the mucosa of the vocal folds and lead to lesions in the posterior larynx.

Choose light foods, and fruits and vegetables chewed thoroughly: besides supplying good nutrients, they relax the muscles of the jaw, tongue, and pharynx, improving diction and leaving the body feeling light.

Avoid chocolate, fried foods, and highly spiced dishes before intensive voice use, as they promote reflux and can increase throat clearing and the feeling of phlegm in the throat. Nor should you lie down after meals; wait 2½ to 3 hours before doing so.

RESEARCH UPDATE · MILK AND DAIRY

For a long time, singers were told to avoid milk and dairy products before singing, on the assumption that they increase mucus production. The available studies, however, do not confirm this effect in healthy adults. In a classic study, volunteers challenged with rhinovirus did not produce more nasal secretions or experience more congestion when they consumed more milk and dairy; those who believed that "milk makes mucus," however, reported more coughing and congestion, without actually producing more secretions (PINNOCK et al., 1990; WÜTHRICH et al., 2005). In practice: if you notice that your saliva becomes thicker after drinking milk, it is worth avoiding it before singing or speaking in public; if you do not, there is no reason for a blanket restriction.

Hard candies, lozenges, and throat sprays may ease unpleasant sensations, but they end up masking the pain of vocal strain, further harming the condition of the mucous membranes.

Apples, with their astringent properties, are recommended for cleansing the mouth and pharynx. Citrus juices, particularly orange and lemon, help absorb excess secretions, but they should not be consumed in excess, to avoid triggering gastroesophageal reflux.

Ice-cold food and drinks can be harmful, because the thermal shock triggers an immediate discharge of mucus and swelling of the vocal folds; some people, however, are more sensitive than others, and this should be taken into account. So hold the first sips of anything ice-cold in your mouth for a few seconds before swallowing, to avoid an abrupt change in temperature at the vocal folds.

Clothing

Clothing can interfere with voice production in three negative ways: through compression, through allergies, and by encouraging poor posture.

Compression around the neck and abdomen causes vocal tension, so choose light, loose-fitting clothes that allow the body to move freely. Avoid accessories around the neck and waist.

Some people are allergic to certain fibers, such as wool. Laundry soaps and fabric softeners can also trigger allergic skin reactions that extend to the respiratory mucosa.

Poor body posture also encourages faulty voice production. For this reason, shoes should ideally be low-heeled and made of natural materials, such as leather. High heels force a tense posture to keep the body upright and consequently stiffen the whole body, particularly the diaphragm, making phonation more difficult.

Sports

Some sports are more beneficial to voice production than others. Swimming and walking are highly recommended: they stimulate circulation and the body as a whole and improve breathing. The activities to avoid are those that demand violent movements and concentrate muscular tension in the neck, back, shoulders, and chest, which ends up in-

creasing tension around the larynx and encouraging a compressed, strained voice. These include boxing, tennis, volleyball, and weight training.

Hormonal changes

During menopause, the decline in female hormones may lower the pitch of a woman's voice. Older men, on the other hand, tend to show a rise in fundamental frequency. People who are physically fit, or whose voices are trained, tend to keep their voices youthful and healthy.

Voice disorders may also recur in the premenstrual period, during the first days of menstruation, during pregnancy, and with the use of oral contraceptives. Birth-control pills usually contain a high dose of progesterone and, although they generally do not have a masculinizing effect on the voice, some women report temporary vocal changes while taking them.

Women who experience premenstrual tension are more likely to show vocal signs linked to the menstrual cycle, such as slight hoarseness, a deeper voice, vocal fatigue, or loss of vocal power, as a result of swelling and hormonal changes.

In late pregnancy, or in the first days after childbirth, breathing is often poorly coordinated. The voice may be lower, hoarse, and tired, with shorter phonation times. Women who use their voices professionally may experience significant vocal limitations during this period.

Lack of rest

Setting the vocal folds into vibration to produce speech requires a great deal of energy, and vocal fatigue can follow excessive or very loud voice use. The fatigue can be so intense that it extends to the whole body.

Usually, after a good night's sleep, the symptoms of vocal fatigue disappear and the voice returns to normal the next day. A poor night's sleep can leave the voice hoarse, weak, and breathy in the morning. Adequate rest is recommended, especially when heavy voice use is planned for the following day.

Beyond physical rest, we must also consider vocal rest. After intensive voice use, the ideal is a period of rest or limited use lasting as long as the voice was in use. Relative vocal rest is recommended, meaning that you limit how much you speak and avoid speaking loudly.



II

PART II

Foundations of Singing

Terminology, voice classification, styles, breathing, and the physiology that sustains the singing voice.

06

CHAPTER 6

The Vocabulary of Singing

This chapter introduces the terms used in professional singing. Understanding them is the first step toward real command of your performance.

Music

The harmonic, melodic and rhythmic combination of musical sounds.

Melody

A combination of successive sounds, heard one after another, as in a scale.

Register

Register is directly linked to the resonance of the sound. Low and middle notes sound most natural with chest resonance; in the upper range,

resonance is felt predominantly in the face and head.

The singer's goal is a homogeneous voice, one that unites the low, middle and high registers into a single color and sound. In vocal training it is essential to blend two or more registers so that the transitions between them become imperceptible. This is achieved through scales and proper breath support, together with a deep understanding of how to stabilize the column of air. Vowels gradually modify as the voice moves through its range, and they play a fundamental role in keeping the larynx in a stable position, balancing low, middle and high notes and bridging the registers.

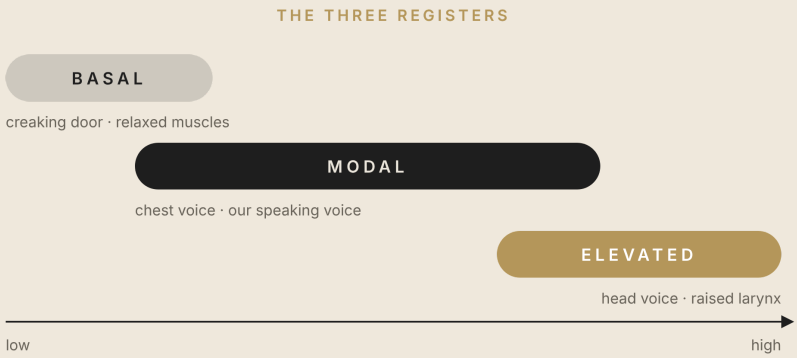
Although some voice scientists prefer to speak of registers and sub-registers, this book works with three vocal registers: **modal, basal (pulse) and elevated**. Some theorists divide the modal register into chest and head sub-registers; here, head voice is treated as part of the elevated register. Within it, head voice corresponds to what we mistakenly call falsetto, a word that implies a "false" voice, when in fact there is no such thing as a false voice.

Modal register. The larynx sits low and the vocal folds vibrate more slowly, setting the bones of the chest into vibration; hence the term "chest voice." This is our natural voice, the one we use for speaking.

Basal (pulse) register. It resembles the creak of a door swinging open. It can serve as an indicator of relaxed musculature (vocal fry, or the basal-sound technique).

Elevated register. Its hallmarks are a raised larynx and faster vibration of the vocal folds. It is also characterized "by energy transmitted in the form of vibratory sensations in the oral, pharyngeal and nasal cavities, that is, in the cranial vault" (DINVILLE, 1993).

The three registers



The three vocal registers across the vocal range: basal (lowest), modal (speech and chest voice) and elevated (head voice). The boundaries vary from person to person and overlap. (TITZE, 1994)

Range

Vocal range is unique to each individual: every person has a lowest and a highest note of their own. Range can be extended, both downward and upward, but doing so requires a great deal of work on the voice.

Tessitura

Essentially, the notes that occur most often in a passage of music and that lie in a more central region of the voice, one that is comfortable and natural to sing in. The more you sing within a comfortable region, the healthier it is for your voice.

Timbre

Timbre is the personality of the voice. It is bound up with harmonics: it results from the many harmonic sounds that make up each voice. The richer a voice is in harmonics, the more beautiful it sounds. Timbre is the control of “color” that lets a voice sound bright, dark, open, closed, nasal, oral, or more or less projected. It is the result of acoustic phenomena that take place in the supralaryngeal cavities.

By modifying the volume and muscle tone of these areas, as well as those of the lips and cheeks, the singer can voluntarily enrich or impoverish the fundamental sound generated by the vocal folds in the larynx. This depends on the order, number and intensity of the harmonics that accompany it, which are filtered in these cavities according to pitch and vowel.

Timbre is the quality of the voice, its identity. It is what allows us to tell one person from another: the quality and distinctiveness of the sound, the particular shape of its sound wave.

Intensity

Vocal intensity is the property that makes a sound louder or softer. Training this dynamic can give the listener an engaging sense of variety. It refers to control of the loudness of the voice, from fortissimo to pianissimo, and to every dynamic gradation in between. It depends on subglottal pressure (the abdominal support that makes vocal power possible).

Intensity is experienced as a sensation of muscle tone distributed throughout the vocal organs. It is perceived as energy transmitted, little by little, to all the resonating cavities and to the muscles of the pharynx

and larynx. As they work, singers must be aware of the muscular effort that intensity demands and of the appropriate, whole-body vocal dynamic that enriches the sound spectrum. Intensity increases with vocal muscle tone and is associated with pitch.

Duration

A sound may last as long as the music calls for, whether short or sustained. Duration is the length of time a sound is produced.

Rhythm

The combination of the durations of sounds and rests within a piece of music (time signatures, tempo): a sequence of sounds organized around a pulse. Differences in style (samba, waltz, tango, rock and so on) are revealed above all by rhythm.

Tempo

The speed at which a song is performed, faster or slower.

Interpretation

Written music includes dynamic markings. In popular music, the arranger notates the dynamics the song should convey; quite often, however, it is the performer who brings his or her own variations, always guided by good taste and good judgment.

Pitch

The quality that makes a sound low, high or in between. It involves controlling the modulation of frequencies, extending the vocal range and singing in tune. To change pitch, one must change expiratory pressure, that is, modulate the degree of abdominal muscle tone, as well as the volume of the supralaryngeal cavities. These adjustments in turn alter the position of the larynx, glottal closure, the vibration frequency of the vocal folds and the place where vibration is felt.

Homogeneity

An essential quality that depends on how the zones of resonance are distributed and on how the voice's different sonorities blend, given their constant interaction. It can be achieved only through the progressive harmonization of all the organs essential to phonation, in other words, through a system of compensation across the entire range and on every vowel, so as to soften or reinforce certain formants (the frequencies that help determine timbre).

Intonation

Intonation is the counterpart of homogeneity. It depends on well-distributed pressure and muscle tone, which ensure proper adduction of the vocal folds and the appropriate shaping of the resonating cavities. Intonation is governed by extremely delicate movements and by mastery of a set of sensations to which the singer must pay very close attention.

Some singers sing flat because they fail to sustain the breath, owing to low muscle tone; others sing sharp, either from excessive pressure or because their breath is spent too quickly.

Food for Thought

> *Whatever the changes in timbre, intensity or pitch, and whatever the changes in the surrounding conditions, vocal technique must remain the same. The singer need only adapt it to these different parameters.*

From a functional standpoint, the carrying power of the voice is always related to the energy expended. It shows itself chiefly in an awareness of muscle tone throughout the whole body, in sharper proprioceptive sensations and in a richer acoustic play of timbre.

Unlike other instruments, the voice has no physical form one can see or touch, and for that reason it is harder to control. This is why singers must have at their disposal a secure and conscious technique, grounded in precise sensations and movements, so that they never lose command of their voice, wherever they happen to be singing.

We can say, then, that the essential condition for a voice that carries is control of internal sensations. It is above all the muscular sensations that tell us what the organs are doing. The work of the muscles, the movements of the organs and the awareness of vibratory tremor are all linked to the propagation of the sound wave through the pharyngeal cavity, along the musculomembranous wall that runs in front of the spine. This is the “place of the voice,” as singers call it: a place they seek out with the help of humming, to arrive at a sound and a vibratory sensation that is felt differently depending on how the larynx moves, the harmonic richness of the laryngeal sound and the pitch. Listeners do not perceive these sensations, but they are essential to the singer. After long training, they become an effective means of control.

These sensations must be complemented by auditory-phonatory control, that is, by listening to one's own voice, since acoustic output is monitored by the inner ear, which makes it possible to evaluate and appreciate the quality of the sounds produced. The singer thus draws, on the one hand, on the ability to control the voice heard through the air and, on the other, on a set of oral, pharyngeal, nasal and laryngeal vibrations that reach the inner ear through bone conduction. For low notes, these vibrations travel by way of the muscles that lower the larynx; for high notes, by way of the muscles that raise it.

Auditory sensations vary with the venue and its acoustics. Sometimes the singer feels the voice is ringing too much; at other times he or she can barely hear it, and the voice seems muffled. Singers who do not want to force their voices should therefore attend to their internal sensations, both muscular and vibratory, because these do not change and do not depend on where they happen to be.

The singing voice is acquired with the help of these sensations, which arise from complex postures and relationships. Becoming conscious of them is indispensable to vocal quality and stability.

Beginners need sensory and vibratory information, delivered through exercises and complementary procedures that build the muscular coordination acting directly on breathing and, at the same time, on the vocal and articulatory organs. To make it easier to discover these different sensations, teachers use a vocabulary of imagery, metaphor and subjective evocation, together with a mental picture of the sound, its color and its mechanism, all of which indirectly bring about the right posture for phonation. The aim is to find a balance between expiratory pressure and vocal emission, expressed as a dynamic engagement of the whole body.

“

Timbre is the personality of the voice.

CHAPTER 6 · THE VOCABULARY OF SINGING

07

CHAPTER 7

Voice Classification

Voice classification should rest primarily on anatomical, morphological and acoustic grounds. Several factors must be considered, some of them predominant and others secondary. Without a careful assessment, one that takes time and patient observation, singers run the risk of singing in a tessitura that does not suit them. Even if they find it beautiful and enjoyable to sing with a particular timbre, the consequences for their voice can be disastrous.

A voice is as individual as a fingerprint. Many singers enjoy success for a while and then develop a range of vocal symptoms, such as fatigue and hoarseness, that can become chronic and pull them away from their careers. The problem is that once an artist has become successful in a musical style that is not necessarily the best fit for his or her voice, changing that public image is extremely difficult.

PREDOMINANT FACTORS

Tessitura	The set of notes the singer can produce easily (notes sung comfortably).
Vocal range	The full set of notes, from lowest to highest, that the singer can produce (notes sung both with and without comfort).

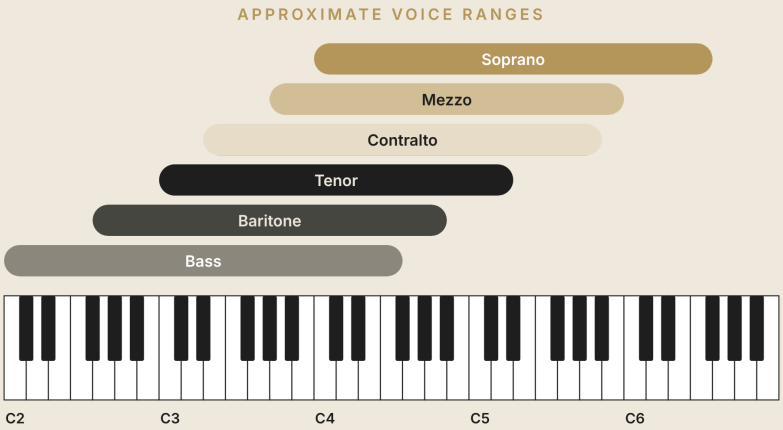
SECONDARY FACTORS

Respiratory capacity and the development of the chest and abdomen.
The pitch of the speaking voice, provided the person uses the pitch that matches his or her anatomical makeup.
Vocal amplitude, which indicates a voice with full, rounded sonorities across the entire range.
Intensity, which allows for power without strain.
Temperament, which represents the singer's overall qualities in relation to his or her vocal possibilities.
Morphological characteristics. It is commonly held that tenors and sopranos tend to be short, stocky and full-figured, and that contraltos and basses tend to be slender, but there are exceptions.

Voice Ranges

FEMALE VOICES		MALE VOICES	
Soprano	C4 – F6	Tenor	C3 – D5
Mezzo	A3 – B5	Baritone	G2 – A4
Contralto	E3 – A5	Bass	C2 – F4

Pitches are given in scientific pitch notation (middle C = C4).



Approximate range of each voice type shown on the keyboard. These bands are teaching guides: every voice has its own limits.

“

A vocal signature is as unique as a fingerprint.

CHAPTER 7 · VOICE CLASSIFICATION



08

CHAPTER 8

Singing Styles

Broadly speaking, singing falls into two large families: classical singing and popular singing. Within each, we find a variety of styles.

Classical Singing

Over the course of history, classical singing has been approached in many ways, each with its own language. Yet ever since the fourth century, when chant emerged in the churches of Rome, the priorities have remained the same: mastery of the coordination between breathing, phonation and articulation; register transitions with no perceptible change in timbre; and flawless vocal agility.

Italian opera was an original art form that grew so powerful and important that the voice itself came to be cultivated intensively, and a technique was developed to perfect it. The Italian language, rich in pure vowels and with few final consonants, lent itself naturally to singing.

The castrated singers (*castrati*), who rose to fame in the sixteenth century, deserve mention here. Their renown spread across the European continent. The practice consisted of castrating boys at a very young

age and subjecting them to relentless vocal training. The *castrati* were considered voices without equal. Their sound was the acoustic result of a small, boyish larynx, left undeveloped by the disruption of male sex hormones, whose fundamental frequency, generated by the vocal folds, was reinforced by the resonators of an adult body. The rest of their growth proceeded normally under the control of the pituitary gland.

Still within the classical tradition, recitative, a declaimed style of song, came to prominence in 1602; physiologically, its production resembles the singing voice. From there *bel canto* gradually emerged, demanding perfect technical mastery, great beauty of tone, an extended range and remarkable agility. Great volume was of little interest, since performances generally took place in palace salons, which were not as large as the theaters built later.

Reactions against *bel canto* began to appear, and the expressive power of the orchestra was gradually consolidated, bringing innovation to the art of instrumentation.

In the Romantic period, the voice took on great importance in expression, declamation and the portrayal of characters. Thus, in early nineteenth-century Paris, the covered, supported voice emerged, setting aside the falsetto and the gentle voice so prevalent in *bel canto*. These were replaced by genuine full-voiced top notes, sung by voices that brought virile, convincing characters to life.

German, French and Italian opera, and the Romantics as a whole, came to demand voices of great power, expressiveness, range and command of register. This is classical, operatic singing.

In the twentieth century, composers emerged with a style that called for singers who could not only sing but also act, dance and improvise. Along this path, *belting* steadily gained ground. Today we can enjoy it in the great flourishing of musical theater, including popular songs that

“blend” certain elements of classical placement while bringing the resonance forward into the “mask” and producing a grittier, edgier sound (SUNDBERG; GRAMMING; LOVETRI, 1993; BJORKNER, 2008).

Popular Singing

When we compare the two kinds of singing, popular and classical, several important aspects come into play: range, articulation, interpretation, voice classification, quality of vibrato and intensity.

Popular singing does not demand as much range as classical singing, in which compositions explore the extremes of the voice. In popular singing, what prevails is freedom of expression, without such strict technical rigor. Thanks to the microphone, popular singers can use the voice at lower intensity and, for example, interpret a song in a personal way, varying rhythmic phrasing and improvising melodically.

Another aspect is the way the musical phrase is pronounced. In classical singing the sound comes first, and every sacrifice is made to beautify it, even at the expense of the intelligibility of the words and lyrics, which are of paramount importance in popular singing.

Genres in Popular Music

Popular music encompasses countless genres, such as samba; choro-inho; frevo and baião, both from Brazil’s Northeast; maracatu; música caipira (Brazilian country music); the nativista music of Rio Grande do Sul; traditional ballads; rock; blues; and jazz. In general, these genres are tied to rhythm, to culture and to the shaping of a society.

Popular Music in the Twentieth Century

At the turn of the twentieth century, popular music took the form of light opera, or "operetta," and of the entertainment that dominated the theaters. Fifty years later, the world had fallen under the spell of rock 'n' roll. The catalyst for this transformation came when the descendants of enslaved West Africans gave the United States the rhythm of jazz.

In turn-of-the-century Brazil, composers in Rio de Janeiro began playing foreign rhythms in a very distinctive way. The instruments they used lent the music a melancholy tone, which gave this new style its name: choro (literally "weeping"), an intricate, largely instrumental genre. The first piece to bear the name was "Apanhei-te, Cavaquinho," by Ernesto Nazaré. Another pioneer was Chiquinha Gonzaga, who also enjoyed enormous success with her marches and modinhas (sentimental parlor songs), including the celebrated "Abre Alas," considered the first Carnival song written in Rio.

Musical comedy developed alongside jazz. Broadway, Manhattan's theater district, reached its peak in the 1920s. Memorable songs captured the public imagination in lighthearted shows such as *Lady, Be Good* (1924), George Gershwin's first major musical, which fused classical, popular and jazz idioms to create a new, quintessentially American sound.

In Brazil, choro was no longer purely instrumental; it was now sung as well. Influenced by maxixe and samba, it began to be played at a faster, more cheerful pace. Thus was born chorinho, or samba-choro, a style that spread through the dance halls of Rio. One of the genre's founders was the composer Pixinguinha.

In the United States, musicals took over the Hollywood screen. Radio star Bing Crosby showed off his distinctive voice in *King of Jazz* (1930).

The end of Prohibition forced many musicians to trade nightlife for daylight. Jazz adapted its style, shaping itself to ballroom dancing in the form of big orchestras and swing. Bandleaders Count Basie, Duke Ellington and Paul Whiteman led the change. Billie Holiday was the great diva of the decade, with her interpretations of popular songs. In 1937, Glenn Miller came to define the era of the *big bands*.

The 1930s were the "golden age" of Brazilian music. Noel Rosa turned everyday situations into poetry in his sambas, as in "Com que roupa eu vou?" ("What Shall I Wear?"), from 1931. With the growth of radio came singers such as Francisco Alves and Orlando Silva, "the singer of the multitudes," who performed sambas by Lamartine Babo, João de Barro and Ataulfo Alves. In 1938, Carmen Miranda sang Dorival Caymmi's "O que é que a baiana tem?" ("What Does the Bahian Girl Have?") in the film *Banana da Terra*. Ary Barroso wrote "Aquarela do Brasil," recorded in 1939 and known to English-speaking audiences simply as "Brazil."

Hollywood continued to launch American stars such as Ella Fitzgerald and Frank Sinatra. In 1942, the musical genius of Irving Berlin produced "White Christmas"; Bing Crosby's version sold more than 30 million records. In contrast to the smoothness of swing, a new style of jazz arose: *bebop*, or "hot jazz," an experimental form built on dissonant rhythms and harmonies, led by saxophonist Charlie Parker, trumpeter Dizzy Gillespie and guitarist Charlie Christian, the first to play with electric amplification.

In Brazil, baião, a genre rooted in the folklore of the Northeast, spread through the cities. The song that shares its name with the genre, released in 1946, was followed by other hits by Luiz Gonzaga, such as "Asa Branca" and "Assum Preto." In 1947, Lupicínio Rodrigues's "Nervos de Aço" ("Nerves of Steel") opened a long gallery of famous sambas-canção, slow, torch-song sambas centered on the theme of dor de cotovelo, the ache of a broken heart.

The blend of Black *rhythm and blues* with white *country music* produced the style that would dominate the decades to come: rock 'n' roll.

Samba-canção remained a staple of Brazilian radio in the 1950s, in the voices of Dalva de Oliveira, Emilinha Borba and Cauby Peixoto. In 1958, a jazz-influenced style emerged that revolutionized urban samba: bossa nova (literally "new trend"). Its leading figures were Tom Jobim and João Gilberto. Gilberto broke with the dramatic, *bel canto*-style delivery of the radio singers, offering sophisticated, intimate interpretations accompanied by acoustic guitar, as in "Chega de Saudade."

In the United States, rock 'n' roll gave way to a revival of *folk* and pure *rhythm and blues*. Singers Bob Dylan and Joan Baez led the movement. Dylan's "Blowin' in the Wind" became an anthem of the civil rights movement.

In Britain, the Beatles created a musical style all their own. In 1962, the single "Love Me Do" launched an unbroken run of chart-topping hits that lasted until 1970. Rock continued to evolve in Britain with the experimental guitar of Jimi Hendrix and the Rolling Stones' mix of *rhythm and blues* and rock 'n' roll. In 1968, a heavier style of rock appeared in the music of Black Sabbath.

In Brazil, the transgressive performances of Caetano Veloso, Gilberto Gil and Os Mutantes marked the birth of Tropicália at TV Record's 1967 song festival; first prize, however, went to "Ponteio," by Edu Lobo and Capinam. Far from the politically charged atmosphere of the festivals, young people in São Paulo were having fun with rock 'n' roll, creating the Jovem Guarda ("Young Guard") movement, whose leading names were Erasmo Carlos, Wanderléa and Roberto Carlos. The 1960s also saw a return to traditional samba, with recordings of songs by Cartola and Nelson Cavaquinho in the voice of Nana Caymmi.

Led Zeppelin's style of rock became known as *heavy metal*, and in the 1970s new bands such as Van Halen (1974) and Def Leppard (1977) adopted the new sound. A blend of rock and melodic pop gave rise to the *glam rock* of Elton John and David Bowie. In 1976, the Sex Pistols assailed Britain with "Anarchy in the UK," ushering in punk rock.

Disco, by contrast, gained international momentum, with a pop beat made for dancing and an emphasis on studio production. The elaborate vocal harmonies of the Swedish group ABBA were a hit all over Europe, with songs such as "Dancing Queen" (1976). In Jamaica, ska and rock-steady evolved into reggae, with its strong, heavy syncopated beat and deep bass line. Jamaican singer Bob Marley popularized the genre worldwide with "Get Up, Stand Up," from 1973.

In Brazil, Milton Nascimento released the album *Clube da Esquina* in 1972; two years later, Elis Regina reached the peak of her career when she recorded *Elis & Tom* with Tom Jobim, singing classics such as "Águas de Março" ("Waters of March"). Composers devoted to instrumental music emerged, such as Egberto Gismonti and Hermeto Pascoal. In rock, Rita Lee, having parted ways with Os Mutantes, launched her solo career, and Raul Seixas became famous with the 1973 song "Ouro de Tolo" ("Fool's Gold").

Punk survived through the 1980s in Britain and gave rise to *hardcore*: fast, distorted and uncommercial. In the United States, Madonna's debut album (1983) produced American hits such as "Holiday" and "Lucky Star." Two other vocalists stood out: Prince and Michael Jackson, whose *Thriller* (1982) became the best-selling album of all time.

From the 1990s onward, popular music became even more globalized and diverse. Singers such as Whitney Houston and Mariah Carey established a heavily melismatic style of singing in pop and R&B that has influenced generations of performers, a subject we will return to in the chapter on melismas. Hip-hop and R&B came to dominate the charts; sertanejo, Brazil's country music, took on a fresh look in the voices and arrangements of so-called sertanejo universitário ("college sertanejo"); funk carioca, the electronic dance music of Rio's favelas, became one of the most listened-to genres in the country; and with the internet and *streaming*, styles from all over the world, from K-pop to trap, now sit side by side on the same playlists. For singers and voice teachers alike, this means an ever more diverse repertoire. A command of the funda-

mentals of technique, vocal health and ornamentation is what allows a singer to move between styles without compromising the health of the voice.

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Learning to sing is, first of all, learning to breathe.

CHAPTER 9 · BREATHING AND SUPPORT

09

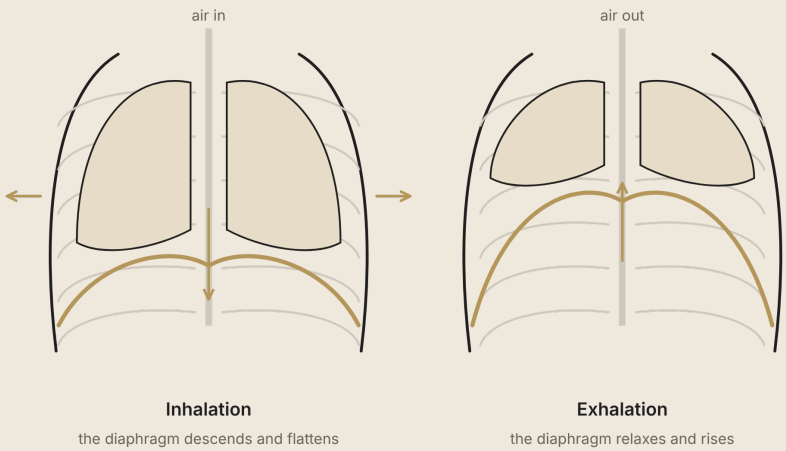
CHAPTER 9

Breathing and Support

The Respiratory System

The respiratory system comprises the thorax, which houses the lungs. The thorax is made up of twelve pairs of ribs, attached at the back to the spine, that curve inward and slope downward. Through their cartilages, the ribs connect to the sternum at the front, with the exception of the last two pairs, the floating ribs, which give the lower part of the thorax greater mobility from front to back and from side to side. The ribs are joined to one another by inspiratory and expiratory muscles. A very powerful set of muscles covers the rib cage and makes possible the movements needed to draw air into the lungs and expel it.

THE DIAPHRAGM: INHALATION AND EXHALATION



On inhalation, the diaphragm descends and the ribs open; on exhalation, the diaphragm relaxes and rises.

The abdominal wall is made up of several muscles, chief among them the transversus abdominis, the external oblique, the internal oblique and the rectus abdominis. Together they give the respiratory system its elasticity and take part, as a whole, in the dynamics of breathing and in supporting the voice. We must also remember the latissimus dorsi, in the lower back, which is used in inhalation and, above all, in exhalation. The work of this powerful musculature is indispensable for keeping the ribs firm during singing. The activity of this group of muscles makes it possible to modulate and regulate expiratory pressure according to the demands of the music, through conscious and continuous control.

Lung capacity varies from person to person, according to age and sex. It also differs with posture: it is smaller when lying down, greatest when standing and slightly reduced when sitting. In the singing voice, the volume of air used normally ranges from 1 to 1.5 liters, depending on the length of the phrase. It also varies with the type of breathing. In upper costal breathing it is minimal, because this is the narrowest part of the

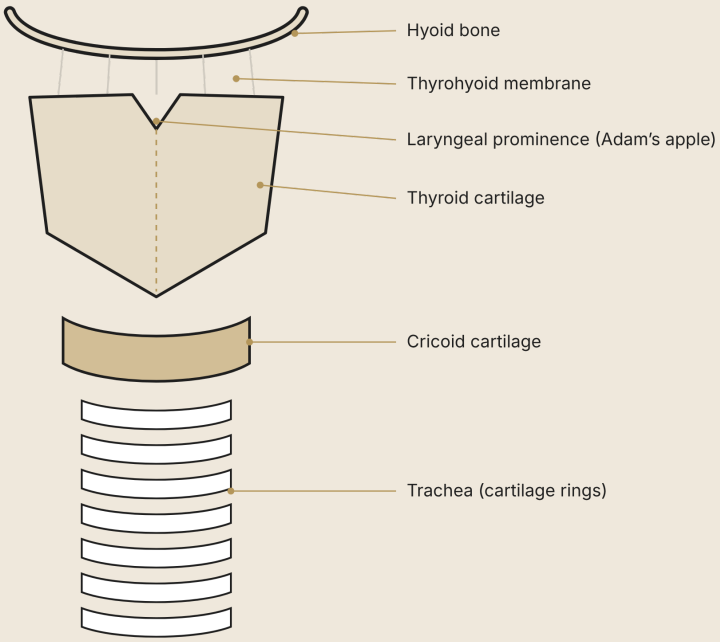
rib cage: it allows little expansion and provokes untimely contractions in the neck and the laryngeal muscles. In addition, the rise of the diaphragm is limited by the absence of abdominal movement.

Breath control matters: the way breath is conserved, disciplined and consciously used to supply the pressure the music requires.

The Larynx

The larynx sits above the trachea, in front of the lower part of the pharynx, behind and below the base of the tongue, at the level of the last four cervical vertebrae. It consists of numerous cartilages linked and articulated with one another, and with neighboring structures, by ligaments and membranes. It is moved by a set of muscles that run in every direction and are covered by mucous membrane.

THE LARYNX, FRONT VIEW



The larynx seen from the front: hyoid bone, thyroid and cricoid cartilages, and the top of the trachea.

Some of these muscles lower or raise the larynx; others are constrictors; still others allow the vocal folds to adduct, abduct and move. The larynx is surrounded by the muscles of the neck, whose excessive contraction can obstruct or compress blood flow through the large vessels. The larynx itself has a rich blood supply.

In singing, as in speech, the larynx is a mobile organ and must be free to rise and fall with the contours of the melodic line. But it can rise, lower or shift forward and backward only through the movements of the tongue and jaw and through changes in the volume of the resonating cavities. This is how the larynx preserves its mobility and the mechanism of the vocal folds. It also changes position according to the posi-

tion of the head, expiratory pressure and articulatory posture, all of which result from a physiological mechanism governed by precise movements under our voluntary control.

Through particular kinds of training, the larynx can also be made to adapt in ways other than those just described, or to adopt predetermined, automatic postures that run counter to its physiological function. If we force the larynx into a constantly low position (and therefore one that is tense throughout the entire range), or into a position that is always very high, there will be no relationship between the position of the organ and the pitch being sung. In both cases, these unphysiological positions prevent the resonating cavities from adapting, alter the vibration of the vocal folds, interfere with articulation and distort the timbre. Whenever certain basic principles are disregarded, singing soon becomes difficult.

The larynx is the vibrating element, since it houses the vocal folds. There are two of them, white in color, formed by horizontally positioned muscular folds. Their front attachment lies at the level of the Adam's apple. Their size varies with sex, age and voice type. They are attached to cartilages that are linked to one another by the muscles that make them work, and they are covered by a mucosa set in motion by wave-like movements traveling from bottom to top and from front to back.

The vocal folds come together and move apart in movements that correspond to pitch. The higher the frequency, the higher the sound, and vice versa for low sounds. Their closure, which may be more or less deep and firm, varies in pressure and extent according to the note being sung. They can also lengthen, stretch, thin, thicken or relax.

The vocal folds produce a complex sound, made up of a fundamental, which determines pitch, and the higher harmonics that accompany it, which are multiples of that fundamental. They are set into vibration under the command of the brain and monitored by the hearing centers. At the moment of laryngeal sound production, several conditions must be met: consciously regulated pressure, appropriate shaping of the

supralaryngeal cavities and good placement of resonance. Together, this muscular activity produces and sustains acoustic signals of varying frequency.

The Singer and the Instrument

Singers are their own instrument. They should not shape it to suit their personal taste; the task is simply to learn to use the voice that nature has given them. This happens only when the laws of physics, phonetics, acoustics and physiology are respected. Singers must be able to sing across their entire range, respecting the colors of the voice, with a voice that is in tune and homogeneous, in both loud and soft singing, with no change in timbre and with carrying power that comes without strain.

Singers should rely on visual feedback, which makes it easier to judge the movements of articulation, the tongue and the jaw, and so to avoid untimely tension in the face and in the breathing. This aid is complemented by tactile feedback at the abdomen, the rectus abdominis and the base of the rib cage, which simultaneously reveals faults in breathing mechanics and the quality of the work of the costo-abdominal muscles. It is also possible to feel laryngeal strain by placing the fingers on either side of the Adam's apple, and in the same way to sense when the larynx sits too high or too low relative to its normal position.

Learning to Sing Is Learning to Breathe

The voice exists only from the moment exhaled air from the lungs supplies enough pressure to initiate and sustain the laryngeal sound. Learning to sing is, first of all, learning to breathe, that is, training to develop a way of breathing suited to singing.

Breathing is vital, because it ensures the chemical exchange between air and blood. In that role it is an unconscious, instinctive act whose movements are regular, rhythmic, symmetrical and synchronized. For the student of singing, however, breathing movements must become conscious and disciplined so that, after long training, they grow active and effective and can be placed at the service of the voice and of the music.

This training engages the nervous system and a whole set of muscles, in order to develop the agility and firmness that every muscular activity requires. Breathing for singing differs in some respects from breathing for speech, because of the demands the music imposes.

For singers, then, it is never a matter of letting the breath flow out unchecked. On the contrary, they must learn to conserve and meter it, constantly regulating how much air they spend according to the intensity, pitch, timbre, range and duration of the musical phrase.

Singers must deliberately delay the closing of the ribs, with the help of abdominal support and the strength of the paravertebral muscles. It is the costo-abdominal muscles, anchored on the spine, that regulate expiratory pressure. The muscles of the abdominal wall and of the lower rib cage find their point of resistance against this dorsolumbar musculature.

This costo-abdominal and dorsal firmness is what we call breath management. But this firmness, this muscular activity, must never be excessive. It should be balanced and measured, never give an impression of overexertion, and be ready to release at any moment. Vocal quality does not call for excessive, unbridled pressure, just as muscular strength does not call for rigid muscles; what it requires is muscle tone, the ability to engage or release at any moment, as the singer wishes.

Achieving this takes long years of work, especially at the beginning. Some people are rigid, either because they do not apply themselves to what is asked of them, because they are asked to do more than they

are able to, or simply because it is in their nature. Others, by contrast, quickly acquire an agile action and sufficient capacity, and strike a balance between the work of the abdominal muscles and the opening of the ribs. For them, training will be easy and effective, all the more so if they are attentive and have an instinct for singing.

When phrases are fairly long, the breath must be carefully metered from the outset; how it is metered, however, depends on the phrase and on whether it begins low or high. In the first case (a low start), more air is usually spent, because the vocal folds vibrate more slowly and little pressure is required. Spending too much air prevents the low notes from ringing or tempts the singer to squeeze. The singer must moderate the active contraction of the abdominal wall and work to sustain the breath, extending its duration: the air should be conserved, but not held back. In the second case (a high start), air consumption is moderate and pressure is greater, since the vocal folds vibrate at a faster frequency. This greater pressure calls for more activity from the costo-abdominal muscles and more firmness in the paravertebral region. The closing of the ribs must be delayed.

Between these two extremes lie all the intermediate gradations imposed by variations in pitch, intensity, timbre and duration. At the very top of the range, one should not take in a large quantity of air. It is a common mistake to think that the more air you have, the better supported your high notes will be, but the opposite is true. This confuses capacity with pressure, that is, the quantity of air inhaled with muscle tone. Taking in more air than usual for high notes forces the larynx into exaggerated contraction and also forces the vocal folds to close too forcefully. In singing, you never need to inhale the maximum amount of air; that is reserved for certain breathing exercises.

Because all vocalises move semitone by semitone, from low to high and back again, capacity and pressure must be adjusted constantly. In other words, as you ascend, capacity tends to decrease while pressure gradually increases; the reverse happens as you descend the scale. It

is always a question of metering the amount of air inhaled and the degree of tone imposed on the costo-abdominal and paravertebral muscles.

These adjustments are required by the constant changes in pitch, duration, intensity and timbre, by the position of the larynx and by the vibration frequency of the vocal folds. In legato singing, regulating the breath is even more meticulous and delicate: the flow of air must be controlled, but not held back.

When a sound grows louder, pressure must be increased steadily through a progressive contraction of the abdominal wall and the rectus abdominis. As the sound diminishes, the reserve of air begins to run out; the singer must then maintain pressure to compensate for the dwindling air, with even greater care, because the possibilities for support diminish and singers tend to let their muscles go slack. The activity is less dynamic than in louder sounds, but it must be metered at every moment. The quality of the timbre comes from the muscular work, from breath control and from keeping the place of resonance.

For sounds that need little air, the singer must supply a moderate, equally sustained pressure that builds toward the end of the phrase, especially if the phrase lies high. With well-metered pressure, the sound will keep its vocal color and remain stable. As in legato singing, however, attention must also go to preserving the shape of the supralaryngeal cavities and the place of resonance. The organs must work closely together, with constant concern for effective management of the breath.

When phrases are very intense, a large amount of air is needed, but above all good abdominal support, very low in the body, bracing against the back and paravertebral muscles. At the very top of the range, maximum pressure should never be used: you should always feel that you could give more. Muscular firmness must not be confused with strain or rigidity.

If a sound begins on a vowel, we should not immediately apply all the pressure the note requires, even if it is a high note that calls for a great deal of pressure. You must “step into the place of the sound” with agility and precision. Immediately afterward, pressure should be increased through active, progressive abdominal contraction. This must happen so quickly that no difference in timbre or intonation can be heard.

At the moment of onset, in addition to regulating the breath, the singer must have prepared in advance the resonating organs through which the sound will spread, and must locate in the mind the place where the sound will vibrate. You can recognize a good singer by the way he or she begins a note. Some start with a hard glottal attack; others scoop up to the note from below. Both habits are to be avoided: the first because it traumatizes the vocal folds, the second because it reveals a lack of synchrony between breath pressure and vocal fold vibration.

To carry out this breathing technique, the muscles of the chest and abdomen must be able to supply more or less tone at the singer’s will, even within the span of a single note or musical phrase. Sustaining the breath therefore means being able to adjust pressure at any moment. Controlling the respiratory muscles makes the natural movements of the larynx and of articulation easier and allows the resonating cavities to adapt. The singing voice demands continuous muscular activity, constantly adjusted to every situation. Singers must pay attention to the work of their muscles and remain conscious of their breathing technique, on which the quality of their sound depends.

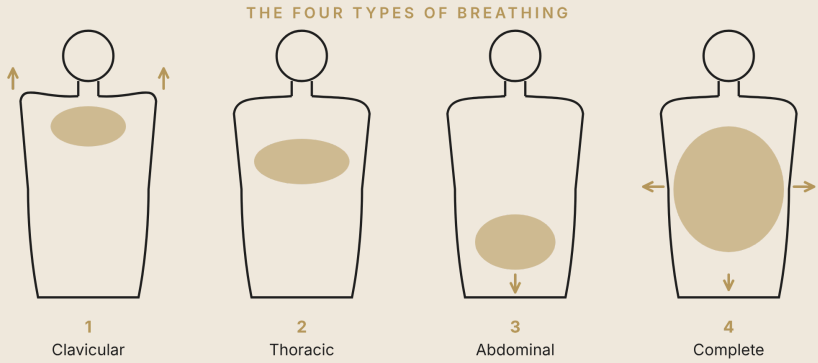
It takes many months before singers can feel and carry out what is asked of them. Beginners who sing easily often see no need to worry about their breath; they would rather “listen to themselves.” When singers trust their own facility and ignore the importance of breathing, serious difficulties may soon appear, such as loss of the high notes, trouble with register transitions and voice breaks, which may in turn be followed by vocal disorders such as nodules.

Given how fragile the voice is, singers must have sound technique, since the voice is not controlled by hearing alone. For that reason, even when they produce beautiful sounds, they need to know what those qualities are owed to. They must also be able to deal with temporary setbacks such as fatigue. In that case, it is better to sing with less intensity and compensate with more resonance, that is, by making the fullest possible use of the resonating cavities. Even for the most gifted, then, regular breathing work aims to train the muscles so that they acquire the agility and tone that every muscular activity requires. Turning an instinctive act into a disciplined, mastered one will allow singers to meet any difficulty they encounter.

Types of Breathing

The literature describes four basic types of breathing:

1. Clavicular, or upper;
2. Middle, mixed or thoracic (chest);
3. Lower, or abdominal;
4. Complete: diaphragmatic-abdominal, or costo-diaphragmatic-abdominal.



The four types of breathing: the highlighted areas show where the greatest expansion occurs in each.

1. CLAVICULAR OR UPPER BREATHING

Clavicular, or upper, breathing is characterized by expansion of the upper rib cage only, which causes a visible lifting of the shoulders and may or may not be accompanied by a forward thrust of the neck. Because lifting the rib cage recruits the neck muscles, one observes contraction of the sternocleidomastoid muscle (which acts as an accessory muscle of inhalation only when one wants to produce a very loud sound) and laryngeal tension, with a narrowing of the thyrohyoid membrane space.

Voice production suffers from the insufficient supply of air, and the resulting sound tends to be high-pitched because the larynx is raised and tense. The sternocleidomastoid muscle is often tense in dysphonic singers, since they tend to keep the rib cage constantly raised.

This kind of breathing focuses on the upper chest. It is not well suited to singing, because air fills mainly the upper portion of the lungs, which is their smallest part; it can also cause dizziness and tingling. As the chest rises, the lower ribs do not open, only the upper ones. These upper ribs are smaller and can open only slightly, because they have little room to expand.

As a result, the singer will have little breath, and unwanted constriction may occur around the vocal folds. It is also hard to keep the spine upright with this type of breathing. It occurs mainly in situations of tension, such as when we are running or frightened, and it is therefore unsuitable for singing, which requires a relaxed neck and relaxed articulators.

2. MIDDLE, MIXED OR THORACIC BREATHING

Middle, mixed or thoracic breathing, the type most commonly observed in the general population, shows little upper or lower movement during inhalation and a forward displacement of the middle chest. It is the breathing we use for most of the day, at rest or in casual conversation, but it is inadequate and insufficient for professional voice use, and especially for singing.

3. LOWER OR ABDOMINAL BREATHING

Lower, or abdominal, breathing is characterized by an absence of movement in the upper region, which is usually underdeveloped, with the shoulders rolled forward, and by expansion of the lower region. It appears in people with little energy, in whom a sense of the chest collapsing is quite evident, but it may also have been cultivated as the "correct way to breathe" as a result of misguided instruction.

This is the breathing used by the German school of classical singing, in which you push the belly outward. This type of breathing pulls the trachea downward, because the diaphragm descends excessively, and consequently leaves the larynx with little mobility. Because the larynx is held very low, the sound becomes darker and the high notes harder to reach. This type of breathing can also make you hunch your back, which makes it difficult to open the lower ribs, the largest ones, and fails to create enough room for the lungs to work freely, pushing the belly outward instead.

The body's balance is reduced, and there is no reason to make this movement, since the lungs are located in the chest, not in the abdomen. If you are a singer who dances, jumps or moves around a great deal on stage, your mobility may be compromised; if your style calls for the extremes of the voice, you will probably be unable to produce healthy distortion. This technique is an old one, and advances in scientific research have shown us more effective ways to breathe for singing. Moreover, with prolonged use of this technique, the belly becomes somewhat slack and grows used to protruding.

4. DIAPHRAGMATIC-ABDOMINAL OR COSTO-DIAPHRAGMATIC-ABDOMINAL BREATHING

Diaphragmatic-abdominal, or costo-diaphragmatic-abdominal, breathing, also called complete or total breathing, is characterized by a harmonious expansion of the entire rib cage, without excess in either the upper or the lower region. The whole lung area is put to use, and it is the most mechanically efficient form of breathing for developing a professional voice. The greater the vocal demand, the deeper complete breathing becomes.

We can therefore conclude that middle, or thoracic, breathing is sufficient for everyday voice use and, in fact, represents the most economical breathing mechanism in terms of energy expenditure.

It is demanding work, because it runs counter to the body's natural movement: when we exhale, the ribs tend to close. Keeping the ribs open in this way creates ample space in the rib cage and lets the lungs work freely, with nothing pressing on them to force the air out.

You are the one in control of your column of air. When you open your ribs, pay close attention to the movement. Do not let your shoulders and chest rise. You will learn to work with the intercostal muscles, which are under voluntary control. That means you can open the ribs alone, without moving anything else, such as the shoulders or neck. It is like opening your hand without moving your arm.

TIP

Always practice in front of a mirror so that you can see your overall posture, and avoid tension in the shoulders and neck.

Support

To sing well and without breaks in the voice, we must use a form of support based on the entire abdominal musculature, including the rectus abdominis, the obliques at the sides, the lower abdominal muscles, the latissimus dorsi and the other back muscles, as well as the pelvic floor. We call this **support**. Support works like a piston that pushes the column of air out of the lungs.

When we use support (the contraction of the abdominal muscles), we push the abdominal organs upward; they in turn push the diaphragm, which pushes the column of air. In this way, the effort is taken away from the neck and shifted to the abdominal wall. The more open your ribs are, the better your support will work.

Good support calls for precision in the use of your energy and physical strength. It is therefore variable: do not overuse it when it is not needed. Some musical genres do not require very strong support, so do not waste your energy for nothing. We will use two terms to describe the variations in air pressure produced by the abdominal muscles:

- **Support:** the act of sustaining. You hold the muscles that keep the ribs open, preventing air from escaping unnecessarily. This keeps the air pressure even as you sing and safeguards the quality of the sound.
- **Counter-support:** pushing the air. Here you use the abdominal and back muscles to control the pressure of the air column according to vocal needs, such as volume, pitch (low notes call for less pressure,

high notes for more), added effects and so on. The abdominal contraction is more vigorous.

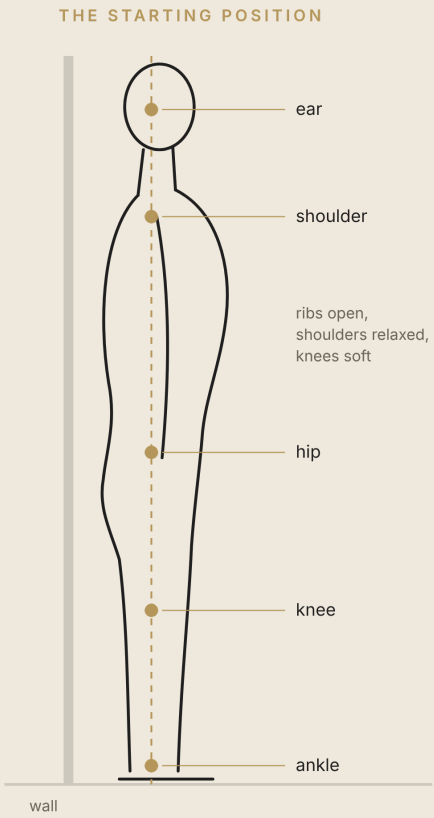
Support is made up of several muscle groups working together:

- the rectus abdominis and the lateral obliques;
- the pelvic floor;
- a complex of muscles in the pelvic region, including the glutes;
- the latissimus dorsi.

Support Exercises

THE STARTING POSITION

All the exercises that follow begin from the same position: standing with your back against a wall, feet hip-width apart, knees soft, spine aligned, shoulders relaxed, chin vertically aligned with your feet and ribs open. Whenever the text asks you to “take the Starting Position,” return to this posture before you begin.



The Starting Position: ear, shoulder, hip, knee and ankle in line, with your back against the wall.

EXERCISE · LEVEL 1 · SUSTAINING

All exercises should be done with your back against the wall.

Take the Starting Position. Inhale, then release the air on a long, gentle "ssss," conserving it for as long as you can. It is worth timing yourself so you can track your progress over the course of your training.

Keep your ribs open: that way you will not compress your lungs or lose air needlessly. Believe me, you may need a reserve of air for longer phrases. Use the rectus abdominis to control the air pressure.

The “ssss” should be continuous, with no variations in intensity. Gradually draw your belly inward.

When you are running out of air, it is time for the insider’s trick. To squeeze out a little more air in songs with extremely long phrases, gradually close your ribs while pulling your belly in and up. This compresses the lungs from the bottom up and lets you use your air to the very end.

Audio for this exercise available for download · see page 154

EXERCISE · LEVEL 1 · STRENGTH: THE FOUR SUPPORTS

1. Take the Starting Position. Inhale and release all your air in five seconds on a very strong “ssss,” pulling your belly in. Notice the pressure produced.
2. Take the Starting Position. Inhale and release all your air in five seconds on a very strong “ssss,” pulling your belly in and pushing your sides out. To feel the side muscles working, press your hands against your waist, just above the hip bones. Cough lightly or laugh heartily: you will feel the muscles move outward. Notice the increase in pressure compared with the previous exercise.
3. Take the Starting Position. Inhale and release all your air in five seconds on a very strong “ssss,” pulling your belly in, pushing your sides out and adding pressure from the pelvic floor. Squeeze your glutes. Notice the increase in pressure compared with the previous exercises.
4. Take the Starting Position. Inhale and release all your air in five seconds on a very strong “ssss,” pulling your belly in, pushing your sides out, engaging the pelvic floor and adding the strength of your limbs to activate the latissimus dorsi. Place your palms together and press them as if you were crushing something between them. Be careful at this point: do not tense your shoulders or neck; the effort stays in the support muscles alone. You can use the strength of your arms and legs to increase the pressure of your support.

Choose the posture that best suits your style and musical genre, the one that feels most comfortable, lets you produce the pressure you want and keeps tension from traveling to the upper body. Notice the increase in pressure compared with the previous exercises.

Audio for this exercise available for download · see page 154

As you will have noticed, each muscle group you add increases the air pressure produced, so you need to learn when and how to use each point of support. The level of pressure depends on the intensity you want, the technique you are applying, the pitch of the note (low and high notes call for different pressures) and the interpretation.

Now that you can control your four supports, practice using them all at once, making sure that no tension is building in your neck. Hold all four supports at the same time, at full strength, and relax your neck. Turn your head to the left and then to the right as you release the air on "ssss," aiming to keep a good level of pressure in the lower body and complete relaxation in the upper body.

EXERCISE · LEVEL 1 · AGILITY

For register transitions, note onsets and fast scales.

1. Take the Starting Position. Practice the consonants /k/ and /g/ in *staccato* (a way of singing in small pulses that produces short, detached sounds separated by brief silences), throwing punches into the air and keeping your ribs open. You will feel the muscles at the pit of the stomach move outward and the lower abdominal muscles move inward. This exercise develops the correct support for phrases that require agility. The abdomen "follows" each note, producing pulses of pressure that sharpen the definition of each sound. Practice for at least five minutes.
2. Take the Starting Position. Inhale and release the air on a *staccato* "s-s-s-s," using the support you learned in the previous exercise. Watch out for hard glottal attacks. If you find it difficult, switch the

“s” to an “f.” Start with one pulse on “s” or “f” per second and, once you are ready, gradually increase the number of pulses per second: two per second, three per second, four per second and so on.

Audio for this exercise available for download · see page 154

General Guidelines

- Always maintain good posture: spine straight, shoulders relaxed (never puff out your chest or raise your shoulders), chin aligned with your feet.
- Continuously observe the behavior of your breathing muscles (abdominal and intercostal).

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The voice exists only when the breath from the lungs provides enough pressure to give birth to the laryngeal sound — and to sustain it.

CHAPTER 9 · BREATHING AND SUPPORT

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CHAPTER 10

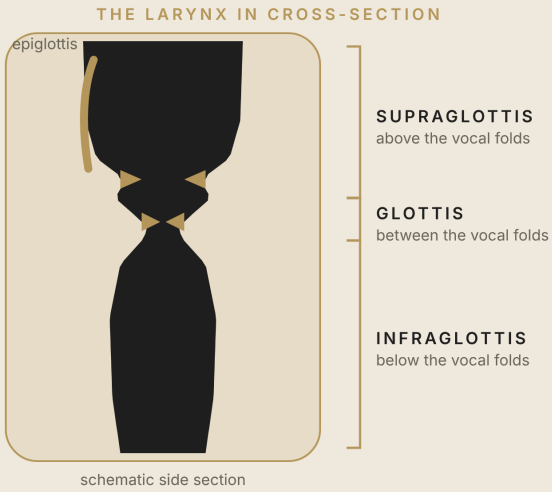
The Physiology of Phonation

What Is the Voice?

Could you say what the voice is? The voice is a sound produced by a complex neurophysiological process. It identifies a person by age, sex, physical characteristics and personality. It also reveals the person's emotional state, and it allows us to communicate.

Where Is the Voice Produced?

The voice is produced in the larynx, a cylindrical organ located in the neck at the level of the sixth and seventh cervical vertebrae, connected below to the trachea and above to the pharynx, what we commonly call the throat. It is made up of cartilage, muscle, membrane and mucosa, and is divided into three portions: the supraglottis, the glottis and the subglottis.



Schematic cross-section of the larynx, showing its three portions: supraglottis, glottis and subglottis.

The supraglottis is formed by the structures above the glottis. The glottis is the space between the vocal folds, where the voice is produced and where it immediately begins to gain resonance within the supraglottis itself. The subglottis is the portion below the glottis.

The larynx has three basic functions: breathing, swallowing and phonation. The last of these is responsible for producing the voice.

Theories of Voice Production

Several theories have been developed to explain how the voice is produced: the vibrating string theory, the mucoundulatory theory, the negative resistance theory, the myoelastic theory, the aerodynamic theory and the myoelastic-aerodynamic theory, which remains the most widely accepted to this day.

This theory describes voice production in terms of the Bernoulli effect: the increase in air velocity as it passes through the larynx draws the vocal folds together, and the elastic action of the muscles makes them

spring back apart. Although it is the most widely accepted theory, it does not account for the full diversity of the human voice.

So how is the voice actually produced? None of the theories of phonation can fully explain such complexity. In what follows, we will try to explain what is known to date about vocal physiology. We know that the voice results from the amplification of phonation in the resonating cavities: the larynx itself, the pharynx, the mouth and the nose.

Phonation itself comes from the air exhaled from the lungs during breathing. As this air passes through the vocal folds, it causes them to come together and vibrate, producing a quiet, weak sound that is subsequently amplified. During inhalation and exhalation without phonation, the vocal folds remain apart.

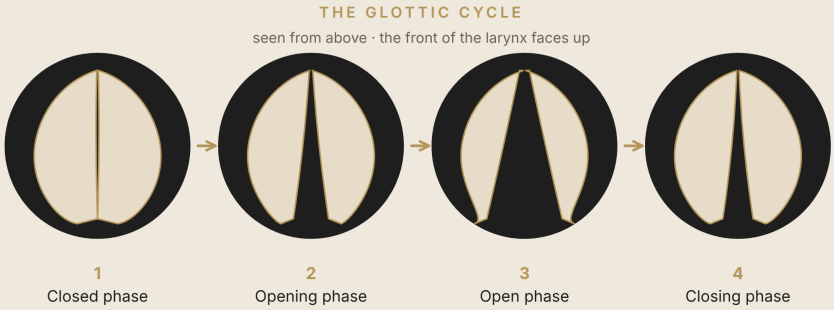
Phonation is far more complex than air passing through and setting the vocal folds into vibration. It is a volitional act, and therefore a movement controlled by the central nervous system. It begins with a stimulus from the cerebral cortex that carries motor information to the larynx, where the movements that produce the voice are carried out. When the nerve impulse reaches the larynx, it activates the lateral cricoarytenoid and arytenoid muscles, which are laryngeal adductors that bring the vocal folds together. Abduction, the process of opening the vocal folds, is carried out by the posterior cricoarytenoid muscles.

The Glottic Cycle

At the same time as the adductor muscles act, the velocity of the exhaled air also increases. As this air passes between the vocal folds, it lowers the pressure in the glottal region, so that the mucosa of the vocal folds is literally sucked inward, bringing the fold on one side into contact with the fold on the other. Because subglottal pressure becomes greater than glottal pressure, the mucosa begins to vibrate. This process is called the **glottic cycle**, and it has four phases:

1. Closed phase
2. Opening phase
3. Open phase
4. Closing phase

These cycles occur at very high speed and cannot be observed individually without stroboscopic equipment.



The four phases of the glottic cycle, seen from above. These cycles can be observed in real time only with stroboscopy.

Vocal Frequency

The frequency of vibration is measured in hertz, the number of cycles per second. The higher the frequency, the higher the sound; the lower the frequency, the lower the sound.

Do you know why women's voices differ from men's? The difference becomes more pronounced during adolescence, through a process called voice change, or mutation, in which changes in the larynx directly affect the characteristics of a person's voice. These changes are driven by new hormone levels and cause the larynx to grow front to

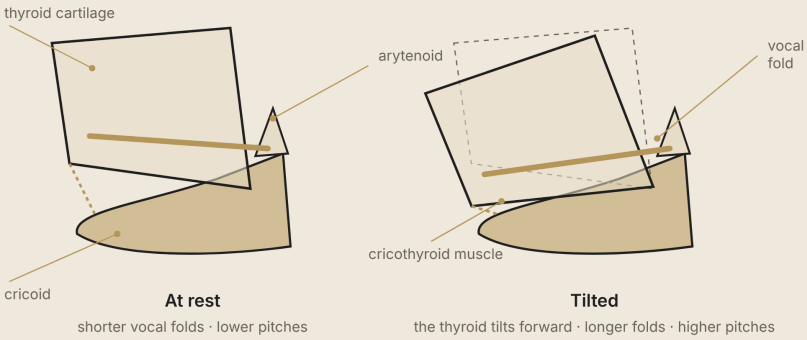
back much more markedly in males than in females. Evidence of this is the protrusion of the thyroid cartilage popularly known as the Adam's apple.

In males, too, the larynx descends further in the neck and the mass of the vocal folds increases, since the greater the mass and length of the vocal folds, the lower the larynx sits and the deeper the voice. The male voice will therefore be lower than the female voice (at least in the most typical cases).

Tension and Elongation

Factors such as the mass, tension and elongation of the vocal folds can lower or raise the frequency of vibration. The last of these, elongation, is chiefly responsible for the range of phonatory frequencies within a single individual; in other words, it is how we are able to make our voice lower or higher.

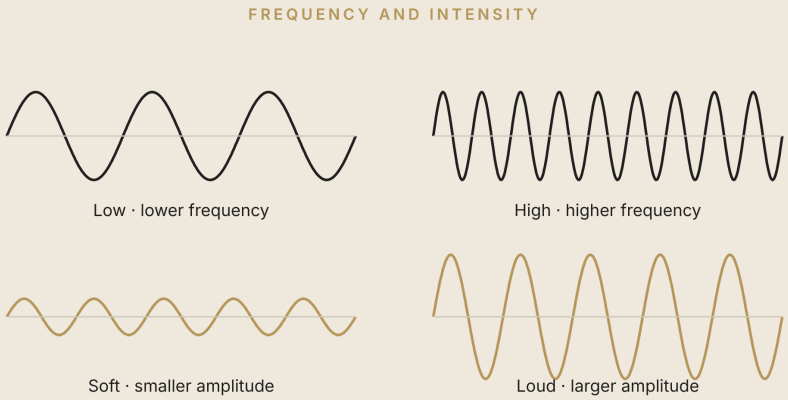
The cricothyroid muscle draws the thyroid and cricoid cartilages toward each other, lengthening the vocal folds and producing higher sounds, in a process known as the **cricothyroid tilt** (the "báscula" movement). The thyroarytenoid muscle, by contrast, shortens the vocal folds, producing lower sounds.



In the cricothyroid tilt, the cricothyroid muscle tips the thyroid cartilage forward and lengthens the vocal folds.

Vocal Intensity

How do we make our voice louder or softer? Vocal intensity is linked to the resistance the glottis offers to the airflow. The greater this resistance, the greater the subglottal pressure and the greater the force of vibration, and therefore the louder the sound. The weaker the resistance, the lower the intensity.



Frequency and intensity: pitch depends on frequency; loudness, on the amplitude of vibration.

Vocal Ontogeny

Why does the voice change over time? The voice develops throughout every stage of life, undergoing changes that accompany a person's physical, psychological and social development. Changes in the size of the larynx, in the thickness and length of the vocal folds and in the characteristics of the laryngeal mucosa and muscles give rise to different vocal configurations over the course of a lifetime. This process is called **vocal ontogeny**.

The Organs of the Vocal and Respiratory Systems

Producing sound involves the movements of the speech organs. To understand the structure and anatomy, as well as the function and physiology, of each organ, and the role each one plays in producing sound, we need to study the vocal apparatus. The so-called phonatory organs serve primarily for breathing, chewing and swallowing; their secondary function is to produce the sounds of human speech.

Lungs

Air is the basic ingredient of speech: it supplies the energy needed for phonation. The airstream used in phonation is set in motion by the walls of the lungs, the intercostal muscles and the diaphragm. Air leaves the lungs, passes through the larynx and exits through the mouth, the nose or both. Speech is normally produced during exhalation, but some sounds are produced during inhalation.

Did you know? On average, the lungs weigh about 700 grams (roughly 1.5 pounds) and measure 25 cm (about 10 inches) in height.

Diaphragm

A broad skeletal (striated) muscle that separates the thoracic cavity from the abdominal cavity. It is attached to the lumbar vertebrae, the lower ribs and the sternum. It has three main openings, which allow the passage of the esophagus, nerves, the aorta, lymphatic vessels and the vessels of the thorax. In humans, this muscle has a ridged appearance and faces upward, with its front part higher than its back part.

At rest, the diaphragm is dome-shaped. During inhalation, the muscle contracts and flattens, increasing the volume of the thorax; in the process, air flows into the lungs to fill the resulting vacuum. When the muscle relaxes, the accumulated air is expelled. The diaphragm also plays an important role in digestion: when it contracts, it exerts pressure on the abdomen. Its movements are likewise important for coughing, sneezing, childbirth and defecation.

Larynx

A short organ that connects the pharynx to the trachea. It lies on the midline of the neck, in front of the fourth, fifth and sixth cervical vertebrae. The wall of the larynx is composed of nine cartilages: three unpaired and three paired. The three unpaired cartilages are the thyroid cartilage, the epiglottis and the cricoid cartilage.

The thyroid cartilage consists of hyaline cartilage and forms the front and side walls of the larynx; it is larger in men because of the influence of hormones during puberty. The back edges of its plates bear short, thick, peg-like extensions called the superior and inferior horns. The cricoid cartilage lies just below the thyroid cartilage, directly above the trachea. The epiglottis is attached to the hyoid bone and the thyroid cartilage; it acts as a kind of “door” to the lungs, through which only air and other gases pass in and out. Liquids and solids do not enter the lungs, because the epiglottis closes and directs them to the esophagus.

The paired cartilages are the arytenoid, corniculate and cuneiform cartilages. The arytenoid cartilage articulates with the cricoid, forming a synovial (diarthrodial) joint. The corniculate cartilage sits atop the arytenoid. The cuneiform cartilage is very small and lies in front of its corresponding corniculate cartilage, connecting each arytenoid to the epiglottis.

The larynx also plays a role in sound production, which results in phonation. On its inner surface is a front-to-back cleft called the laryngeal vestibule, which contains two pairs of folds: the vestibular folds (false vocal folds) and the vocal folds (true vocal folds).

Vocal folds

Two elastic membranes positioned horizontally in the larynx, like a pair of lips, running from the Adam's apple at the front of the larynx (the anterior part) to the arytenoid cartilages at the back (the posterior part). At the front, the vocal folds are attached to the thyroid cartilage, and at the back to the two arytenoid cartilages. Guided by the arytenoids, they can come together, blocking the passage of air, or move apart, letting air pass freely.

The ends of the vocal folds attached to the thyroid always remain together; only the ends attached to the arytenoids move, and it is they that set the vocal folds in motion. The opening between the folds is the glottis, which, when open, is V-shaped.

Epiglottis

A cartilage located at the top of the larynx that acts as its lid or valve, preventing food from entering the trachea and, consequently, the lungs. During phonation and breathing, it opens the entrance to the larynx, leaving the airway clear.

Pharynx

The cavity located between the mouth (the root of the tongue), the nasal passages, the larynx and the esophagus. Air coming from the lungs through the larynx passes through the pharynx and exits through the mouth, the nose or both, depending on the position of the soft palate. The pharynx can change shape and size when the root of the tongue retracts, when the larynx moves up or down, or when the back wall of the pharynx contracts during phonation and swallowing. It acts as a resonator, shaping the resonance of vowel sounds.

The upper portion of the pharynx, called the nasal part or nasopharynx, communicates with the choanae, with the pharyngeal openings of the auditory (Eustachian) tubes and with the oropharynx. The oropharynx communicates with the mouth and serves as a passage for both air and food. The laryngopharynx extends downward from the hyoid bone and connects with the esophagus (the food passage) behind and with the larynx (the air passage) in front; like the oral part of the pharynx, it is both an airway and a digestive pathway. The auditory tube also communicates with the pharynx through its pharyngeal opening, which in turn connects the nasal part of the pharynx with the tympanic cavity of the middle ear.

Uvula

The extension of the soft palate. It is set into vibration when we articulate uvular trills.

Palate

The roof of the mouth. The front part (two-thirds of the palate), bony and fixed, is called the hard palate, or palatal vault. The back part, muscular and mobile, is called the soft palate, or velum. The soft palate can lower, letting air escape through the nasal cavity (as in breathing or in producing nasal sounds). When it is raised and pressed against the back wall of the pharynx, it closes off the nasal passage and air exits through the mouth (as in speaking oral sounds). It can also be partly raised, letting air pass through the mouth and the nose at the same time (as in articulating nasalized sounds).

Tongue

The principal organ of articulation. It finds points of contact on different parts of the palatal vault and can move its tip toward the lower or upper teeth as well as raise its base toward the back of the palate. Through the muscles that connect it to the surrounding structures, it takes part in opening the mouth, in the movements of the soft palate and in the tone and firmness of articulation.

Because all the muscles of this region are directly interconnected, and their fibers run continuously into the muscles of the hypopharynx and larynx, any downward and backward contraction will trigger action in other muscles of the region, whereas a normal position of the base of the tongue eases the contractions of this whole group of muscles. Likewise, any action that seeks to widen the resonating cavities from side to side will tend to ease the action of the muscles that close or narrow the opening of the hypopharynx and the back of the mouth.

Teeth

The teeth serve as a point of articulation. Set in the alveolar ridge, they can partly or completely block the passage of air when they come into contact with the lips or the tip of the tongue.

Nose

A protrusion at the center of the face. Its outer part is called the external nose, and the hollow within it is known as the nasal cavity. The external nose is shaped like a three-sided pyramid with its base at the bottom, whose back face fits vertically against the middle third of the face. The sides of the nose each have a crescent-shaped flare called the ala, or wing, of the nose. The nasal cavity is divided into two com-

partments, one right and one left.

Each compartment has a front opening, the nostril, and a back opening, called the choana. The choanae connect the nasal cavity to the pharynx. It is in the nasal cavity that air is conditioned, that is, filtered, humidified and warmed. On the side wall of the nasal cavity are the nasal conchae (turbinates), divided into superior, middle and inferior.

Trachea

A tube that continues on from the larynx, enters the thorax and ends by dividing into the two main bronchi. It runs along the midline, in front of the esophagus, veering slightly to the right only at its lower end. The framework of the trachea consists of roughly twenty C-shaped rings of cartilage, open at the back, called the tracheal cartilages.

Lips

The lips shape the articulation of contoids (consonant sounds) and vocoids (vowel sounds). In contoids, the lips can partly or completely block the passage of air, as in [f], [p] and [m]. In vocoids, the degree of lip rounding and protrusion distinguishes one sound from another, as in [i] and [ü]. In general, the higher the tongue in the articulation of a rounded vocoid, the more rounded and protruded the lips: in [u], the tongue is high and the lips are more rounded and protruded; in [i], the tongue is at medium height and the lips are less rounded and protruded. The degree to which the lips project forward, like the degree to which the mouth opens, enlarges the resonating chamber and so affects the timbre of the sounds.

Lower jaw (mandible)

The jaw determines how far the mouth opens or closes and the distance between the tongue and the palate, distinguishing high vocoids from low ones and enlarging or reducing the size of the oral cavity.

Resonating chambers

They reinforce and intensify the vibrations (the sounds) produced by the vocal folds. The quality of the sound depends on the shape and volume of the resonator. The resonators are the pharyngeal cavity (pharynx), the oral cavity (mouth), the nasal cavity (nasal passages) and the labial cavity (lips projected forward).



“

Always leave yourself the feeling that you could give a lit-tle more.

11

CHAPTER 11

Exploring Your Own Voice

When the moment comes to sing, singers must prepare themselves physically and mentally. They need to concentrate and form an acoustic image of the sound: its pitch, its color as the ear perceives it, its mechanism and the sensations it produces. This motor and sensory command, which originates in the nerve centers, sets in motion the various movements of the organs and the work of the muscles.

In practical terms, this means the singer assumes a phonatory posture, which requires control of the breathing action and of the amount of air inhaled for the chosen exercise. It is a conscious and indispensable preparation on which vocal production depends: the voluntary shaping of the resonating cavities, the widening of the faucial pillars together with the lifting of the soft palate, and the work of the pharyngeal wall.

Checking the articulatory positions of the tongue and lips for each sound pronounced is essential if the larynx is to carry out its natural movements freely.

From the moment singing begins, a shape, a mold, must be prepared, starting with the low notes. But as the exercises rise semitone by semitone, there is no sudden change in technique: there is continuity, a progression at every level, in expiratory pressure and in the increasing volume of the supralaryngeal cavities, as well as in the movement of the larynx and of the place of resonance, which shift in the same direction.

From one note to the next there is a mode of transmission that is difficult to achieve, one that evolves little by little and must be held in the mind, since this is the only way for the muscular work and the posture of the organs to change progressively with each note. As one descends the scale again, the larynx normally lowers gradually, accompanied by a relaxation of the muscles. If this relaxation is excessive, the voice drops, loses its ring and goes out of tune. To prevent this, singers must maintain the pressure, the way the breath is distributed and the place of resonance. A certain energy must be preserved even as the scale descends, and the upward direction of the breath must not be lost.

It is important to remember that, moving from low to high, the mouth opens progressively, with the jaws separating at the joint where they articulate, together with a diffuse sensation of laryngeal tension and a slight rise in the base of the tongue. As the sound rises, the exact color of the vowel is often lost, because the singer, trying to round out the sound or “yawn” it, has altered the position inside the mouth that corresponds to the vowel being sung. If, on the contrary, that position is respected, the singer obtains the characteristic vowel timbre of each vowel, corresponding to its specific formant regions. Thanks to technique, the extravocalic timbre, the color unique to each of us, is achieved. The different formants are distributed in such a way that, as it

leaves the lips, the voice has an acoustic profile that is truly the singer's own. It is generally accepted that the singing voice involves several formants.

Singers must become aware of the importance of the basic principles of vocal technique and make use of *feedback* from their own voice, since the voice is the product of learned behavior, developed through specific muscular activity and through the different sensations on which this technique rests. It is through progressive, well-guided training that beginners will discover and develop their sensations and learn to master the tone and agility of their muscles. They must also understand that, even in moments of great intensity, they must always be able to do more and remain in command of their muscles.

According to this technique, there must be cohesion, a continuity between the two extremes: the sensation of breath support and the sensation of vibratory tremor. The vocal and respiratory organs are joined to one another in a harmonious, balanced way, never giving the impression of effort concentrated here or there, apart from a very diffuse sensation of taking up the phonatory posture and a well-distributed effort, more or less sustained or firm according to the demands of the music. This set of movements and sensations, sought out systematically, will become, after long training, a global sense of the movements and sensations that make up the technique of the singing voice, and at the same time an effective means of control, valid for every voice.

None of this is easy to achieve. Most of the time it takes long years of progressive work and personal exploration. For students, the point is not to imitate the teacher's voice or to chase a particular timbre! The point is simply to learn that every vocal flaw, and every success, must find its meaning in an awareness of the physical action and in control of the sound. In this way, by using sound technique, students will develop their true voice.

Inappropriate Vocal Behaviors and Habits

- Lack of proper postural alignment between head and body;
- Reduced body movement, with the body held still or moving only minimally while speaking or singing;
- Letting the air out before speaking or singing;
- Taking a deep breath before every utterance;
- Speaking while inhaling;
- Lack of proper coordination between breathing and phonation;
- Keeping the shoulders and chest constantly raised while speaking or singing;
- Generalized tension throughout the body while speaking;
- Generalized tension in the muscles of the vocal apparatus while speaking or singing;
- Visible contraction of the paralaryngeal muscles while speaking or singing;
- Constant hyperextension of the neck;
- Habitually using excessively high intensity;
- Habitually using excessively low intensity;
- Habitually speaking too fast;
- Regularly and actively cheering at sporting events or playing sports that involve using the voice;
- Singing for long periods without proper technique;
- Modulating the voice solely by increasing intensity;
- Fixing the voice on a single pitch;
- Overusing very high pitches;
- Overusing very low pitches;
- Failing to articulate speech sounds distinctly;
- Shifting the jaw sideways instead of opening it;
- Habitually using hard vocal onsets;
- Habitually using breathy vocal onsets;
- Maintaining a constant nasal resonance focus;
- Maintaining a constantly low resonance focus;
- Failing to make use of resonance, resulting in overall hyporesonance;
- Overusing a commanding, authoritative voice;
- Habitually using a whispery vocal quality;
- Habitually using a tense, pressed vocal quality;
- Habitually using the basal register (vocal fry) in everyday speech;
- Habitually engaging in abusive vocal practices, such as forceful throat clearing and coughing;
- Skipping vocal warm-ups and cool-downs;
- Talking over background noise;
- Wearing clothing that is tight around the larynx or the abdomen;

- Habitually imitating harmful vocal models, other voices, animals or noisy objects;
- Talking for long periods with the phone receiver cradled on the shoulder;
- Using a cell phone excessively and/or for long periods;
- Using speakerphone excessively and/or for long periods;
- Not drinking enough water.

Anatomical and Functional Patterns

- Varying intensity by squeezing the laryngeal and paralaryngeal muscles;
- Varying pitch solely by changing the vertical position of the larynx;
- Varying pitch by compressing the glottal and supraglottal structures;
- Limiting jaw movement while speaking;
- Keeping the tongue constantly pushed forward while speaking;
- Keeping the tongue constantly pulled back while speaking;
- Keeping the tongue between the teeth (atypical swallowing, or tongue thrust);
- Holding the larynx in a constantly high position in the neck;
- Holding the larynx in a constantly low position in the neck;
- Using medial (side-to-side) constriction of the laryngeal vestibule to phonate;
- Using excessive front-to-back constriction of the laryngeal vestibule;
- Bulging veins in the neck while speaking.

What to Avoid

- Shouting, or talking for long stretches of time.
- Exposure to sudden changes in temperature, including ice-cold drinks, which can cause an abrupt temperature shift (thermal shock) and trigger a release of mucus. Before swallowing anything ice-cold, hold it in your mouth for a few seconds.
- Alcoholic beverages before singing or heavy speaking, since alcohol acts as an anesthetic: it seems to improve the voice while masking its abuse. The same is true of sprays, lozenges and mint, peppermint or ginger drops.

- Throat clearing or coughing, since these habits create friction between the vocal folds.
- Chocolate drinks and dairy products, especially before professional voice use, since they increase secretions in the vocal tract and lead to excessive throat clearing.
- Overusing the voice in the premenstrual period: during this time a woman's body swells with retained water, and so do the vocal folds, which become heavier and vibrate less easily.
- Exposure to dust, fumes and very strong odors, especially if you have allergies.
- Talking while exercising: during breathing the vocal folds stay open, and during speech they close in order to vibrate. Talking while breathing hard, as physical exercise demands, goes against the physiological nature of the vocal apparatus.

Recommendations

- The pillars of singing (breathing, the use of the resonators and articulation) should at first be the focus of slow, progressive work.
- In the beginning, the best way to build agility is to practice slowly.
- Seek “quality” of voice first; never “force” the voice to work, and always stay relaxed when you sing.
- The body of an instrument does not tense up when it plays. Your body should be loose and free, so avoid muscular tension.
- Studying music, reading music well and being able to sight-sing should come before the study of singing.
- If you have a heavy voice, work extensively with detached, *staccato* sounds and agility exercises. If you have a light voice, work with legato lines and sustained notes.
- At the first sign of a cold, drink orange juice or lemonade, take vitamins and rest. Avoid needless coughing.
- If you feel the need to cough, firm your lips, breathe calmly and swallow.
- Always sing with a straight line in mind, so that the voice does not change “color.”
- Do not sleep with the window open unless you are sure you will sleep with your mouth closed.

III

PART III

Vocal Ornaments

Fry, twang, yodel, vibrato, melismas, cry and appoggiatura: the devices that give the singing voice its identity and expressive power.

Ornaments are what turn a technically healthy voice into a voice with a signature of its own. Before exploring them, it is worth recalling what the previous chapters have shown: nearly all of them depend on breathing, support and relaxation. Practice each technique slowly, without discomfort and, whenever possible, under the guidance of a voice teacher or a speech-language pathologist.



12

CHAPTER 12

Fry

Fry, or *creak* (which speech-language pathologists call vocal crepitation and some linguists call creaky voice), is a low sound made of widely spaced pulses, reminiscent of a creaking door. It differs from sounds produced by air turbulence, such as the sibilant fricatives /s, z, ʃ, ʒ/, whose main source is the airstream striking an obstacle, the teeth (LADEFOGED; MADDIESON, 1996). In fry, the noise comes from the irregular vibratory pattern of the vocal folds themselves. Hollien and Michel (1968) described it as a true phonatory register rather than a pathological change.



Modal
regular, closely spaced pulses

Fry
widely spaced, irregular pulses

In fry, the glottal pulses become widely spaced and irregular.

Relaxed fry is characterized by a lower larynx position; predominant action of the thyroarytenoid (TA) muscle, especially its internal bundle; less involvement of the vestibular (false) folds; looser vocal fold edges within a more relaxed laryngeal framework; single glottal pulses; and low intensity.

Tense fry is characterized by a raised larynx, with a consequent increase in glottal adduction; action of the lateral cricoarytenoid (LCA) and external TA muscles; greater involvement of the vestibular folds; stiff vocal fold edges within a tenser laryngeal framework; double or even triple glottal pulses; and an intensity equal to, or slightly greater than, that of relaxed fry.

In American English, this crackling voice is called *vocal fry*; in British English, *creaky voice*. Both terms tend to be conflated with the pulse (basal) register. The authors stress, however, that the two are not necessarily equivalent: the pulse register is defined by a very low fundamental frequency (f_0) with a predominantly creaky quality, but creak can appear at any frequency in the vocal range, not only in the pulse register. Thus, "the term *vocal fry* refers exclusively to creak in the pulse register, whereas the term *creaky voice* denotes creak introduced into any type of phonation." Pulse-register sound is a vocal quality that often appears in relaxed voices, signaling low tension, or in tenser voices, conveying expressions of "surprise, admiration or suffering."

Tense fry

In clinical practice, tense fry may be recommended in cases of glottal insufficiency caused by unilateral laryngeal paralysis, sulcus vocalis and abductor spasmodic dysphonia, with the aim of improving glottal closure and vocal stability.

13

CHAPTER 13

Twang

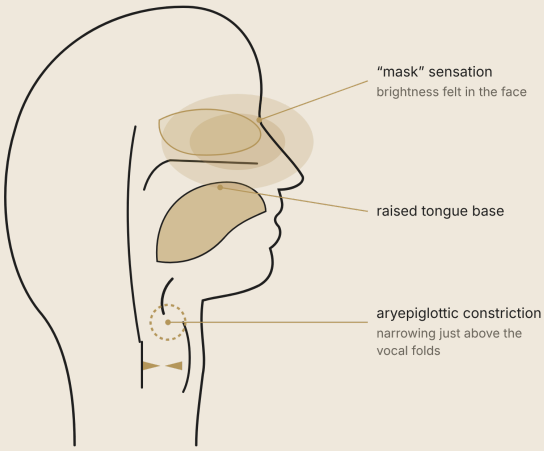
Nasality is one of the most widely studied aspects of voice quality and has been described in phonetics since the seventeenth century. Attempts to explain it have produced a number of oversimplifications. The classic claim is that when the velum closes, speech is not nasal, and conversely, that when speech is not nasal, the velum is closed. Another oversimplification holds that nasal emission always increases nasality and, conversely, that nasality requires nasal airflow. Laver (1980) reminds us that nasality is, in essence, a matter of resonance.

According to Kayes (2004), **twang** is physiologically characterized by aryepiglottic constriction, a high larynx and a raised tongue dorsum. The thyroid cartilage may be tilted or neutral, which allows the vibrating mass of the vocal folds to vary. Aryepiglottic constriction tends to increase resistance at the point of vocal fold closure, so it is important not to push a great deal of air pressure when producing this quality. Twang can also be either nasal or oral.

How difficult it is to perceive and understand the mechanism of vocal resonance became clear in 1940, when Paget conducted an experiment with a resonator tube shaped to produce the vowel /a/. When the tube was squeezed right at its opening, it produced the nasal quality of the American twang, a sound associated with New Englanders and unmistakably characteristic of Western *country* singing. This confirmed his earlier observations that the perceptually nasal quality of twang comes not from nasal resonance but from a constriction in the pharynx (YANAGISAWA et al., 1989).

Twang is associated with "mask resonance" ("singing in the mask"): a bright, metallic sound (*humming* helps add that metallic edge). The base of the tongue may be raised gently, for a subtle boost in metallic quality, or raised high, for a strongly metallic sound. The tongue needs to be well conditioned, almost "worked out," so that it can keep its base raised and spread; its upper sides should rest against the upper molars and never leave them while singing. The tip of the tongue should stay on the floor of the mouth.

TWANG: BRIGHTNESS AND "MASK" RESONANCE



Twang: a raised tongue base, aryepiglottic constriction and a sensation of brightness in the face.

WHERE THE TERM COMES FROM

Twang is one of the voice qualities systematized in the vocal training model developed by the American researcher and teacher Jo Estill, who studied the role of aryepiglottic constriction in producing a "bright" sound (YANAGISAWA et al., 1989; STEINHAUER; KLIMEK; ESTILL, 2017).

LISTEN TO

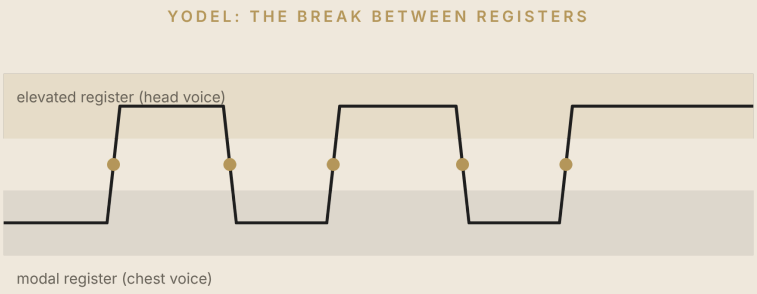
Examples of twang in popular singers: **Christina Aguilera**, **Demi Lovato** and **Bruno Mars**.

14

CHAPTER 14

Yodel

The **yodel** (or *yodelling*) is a singing technique characterized by rapid, deliberate alternation between two registers, usually the modal register (chest voice) and the upper register (head voice). Register shifts are tied to changes in voice quality, and abrupt transitions between registers are usually perceived as more or less noticeable "breaks" in the voice. These breaks can happen involuntarily, but they can also be produced on purpose and even exploited artistically, as in *yodelling* or in certain styles of contemporary music.



The yodel deliberately alternates between the modal and elevated registers.

The yodel and *country* music

Country was the music of everyday rural life among white Southerners in the United States, shaped by the music of the British Isles, by Appalachian music and by the hymns of the Puritan churches, kept alive by descendants of English settlers.

At first, *country* was known as *folk songs* and was later nicknamed “white *blues*,” because its lyrics dealt with rural life in a similar way. Over the course of the genre’s history it was also called *hillbilly* music and, finally, *country music*. White rural laborers were disparagingly called *hillbillies* (roughly, “backwoods folk”), which is why, as cities grew, “white *blues*” came to be known by that name.

One of the genre’s chief pioneers was Jimmie Rodgers, nicknamed “the Singing Brakeman” (MAZZOLENI, 2012). He became a benchmark for the genre thanks to the vocal inflection of his yodel, particularly in “Blue Yodel,” one of the first *country* songs to sell a million copies (MAZZOLENI, 2012). The kinship with the *blues* lies in the fact that these were songs with sad themes and the word “*blue*” in the title, performed with nothing but guitar accompaniment.

From Jimmie Rodgers onward, *country* acquired a distinctive trait that set it apart from Black *blues*: its way of singing. After him, other artists further distinguished themselves from the *blues* by singing in duos, as The Delmore Brothers did, for example.

“

Vibrato gives the voice expressive richness and great emotional power.

CHAPTER 15 · VIBRATO

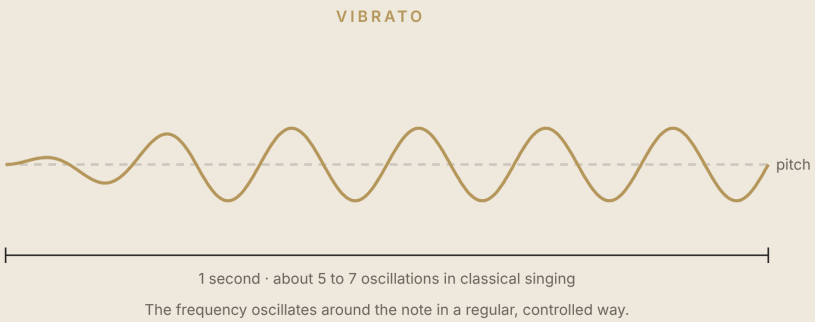
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CHAPTER 15

Vibrato

Vibrato is a regular modulation of the fundamental frequency used by classical singers and by some popular singers. It is a kind of orderly physiological tremor that makes the fundamental frequency rise and fall over a small range several times per second. It is considered a form of tremor, but one whose sound waves are similar or identical in length and amplitude. It is important to remember that vibrato is not the same as a shaky voice, which stems from a lack of training.

The foundation of vibrato is breathing and vocal support. Without proper support, it will be hard to control vibrato and sustain notes; without good breathing, there will be no good vibrato. Ideal support, airflow and adequate resonance allow vibrato to emerge naturally.



Vibrato: a regular oscillation of frequency around the note.

Vibrato is used on phrase endings, on intermediate notes, in melismas, in improvisation or on specific vowels. Any agility technique, such as melismas and the appoggiatura, depends on complete control of the voice, which begins with control of vibrato.

Vibrato gives the voice expressive richness and great emotional power; it is a resource as important as it is complex. Its rate ranges from 2 to 8 Hz, with an average of 5 to 7 Hz observed in classical singers (SUNDBERG, 1987); its intensity varies by 2 to 3 decibels.

Vibrato is not a well-defined phenomenon and has been interpreted in various ways: as an oscillation of the cricothyroid muscles, as an alternating modulation of the cricothyroid and arytenoid muscles, as an alternating contraction of the laryngeal and diaphragmatic muscles, or as a tremor of the respiratory and vocal tract musculature. Vibrato generally diminishes at soft intensities and increases in loud sounds.

The oscillation seen in vibrato may synchronously involve other parts of the vocal tract, such as the jaw, the velum, the base of the tongue, the epiglottis and the lateral and posterior pharyngeal walls. In professionals with sound technique and great natural gifts, vibrato is quite regular and controlled, reflecting a natural coupling between the vocal tract

and the larynx. An aperiodic vibrato, by contrast, usually results from insufficient breath support or forced use of the voice, and may also be related to nervous tension.

A vibrato slower than 5 Hz is considered slow and is used in jazz, *blues* and African American popular music. In Brazilian sertanejo (country) music, one sometimes hears a forced, exaggerated vibrato on practically every sustained vowel.

Curcio, Behlau and Pontes (2000) studied vibrato in three singing styles and found that vibrato rate is similar in opera, rock and sertanejo singing, although stability is greater in opera. This study supports the notion that vibrato is a well-defined neurological phenomenon that can be observed in all individuals.

Narrowband spectrograms display the individual harmonics of a sustained sung vowel, in which the presence of vibrato can be observed.

BEWARE OF OVERUSE
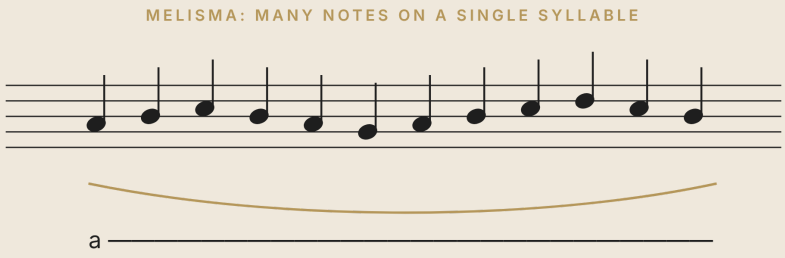
Vibrato is the foundation for other techniques, such as melismas, but it should be used with care. Many people who discover this technique apply it to everything, all the time, which becomes extremely unpleasant for the listener.

16

CHAPTER 16

Melismas

Melismas are popularly known as “runs” or “riffs.” The technique consists of changing the note (the perceived pitch) of a syllable while it is being sung, and it is closely tied to vocal agility.



In a melisma, a single syllable travels through several notes.

Melismas have their roots in Gregorian chant, in which choristers drew syllables out at length. With European colonization, enslaved Africans in the Americas were influenced by Gregorian and classical singing, giving rise to faster, intuitively improvised melismas that, blended into

African American spirituals (*gospel*), acquired a *swing* that propelled and modernized *black music*. The styles of *black music* that use melismas include *gospel*, *R&B*, *soul*, *neo-soul* and *blues*.

The music of ancient cultures used melismatic techniques to induce a hypnotic state in the listener, useful for mystical initiation rites and religious worship. This quality can still be found in Indian and Islamic music.

Melismas are, more often than not, highly intuitive or learned by imitation. But for those who want to start learning them and get a sense of how to stretch a syllable without going off pitch, using the right notes, the way in is to begin by singing syllables over some of the scales typical of *black* styles: the blues pentatonic scale and the major and minor pentatonic scales.

TODAY

In contemporary pop and R&B, melisma is the hallmark of performers such as Whitney Houston, Mariah Carey, Beyoncé and Ariana Grande, and it influences singers all over the world, including in Brazil. That is precisely why it pays to practice carefully: fast, extended melismas demand support, accurate intonation and agility, and they become a source of tension when performed without preparation.

17

CHAPTER 17

Cry

As its name suggests, *cry* draws on the act of crying. The technique consists of knitting the eyebrows, as in a crying expression, in order to achieve:

- greater vocal muscle firmness;
- better quality in the high notes;
- vocal projection;
- a more metallic sound (brilliance in the voice) across vocal functions or modes, even in more neutral phonation.

> WHERE THE TERM COMES FROM SEE THE CRICOTHYROID TILT IN CHAPTER 10.
>
> In Jo Estill's model, *cry* (also called *sob*) is one of the voice qualities described. It is associated with tilting of the thyroid cartilage through the action of the cricothyroid muscle, which helps thin and lengthen the vocal folds (STEINHAUER; KLIMEK; ESTILL, 2017). The "crying" expression serves as a helpful image for reaching this adjustment.

LISTEN TO

Examples of cry in popular singers: **Steven Tyler**, **Xororó**, **Adele**, **Alicia Keys** and **Michael Jackson**.

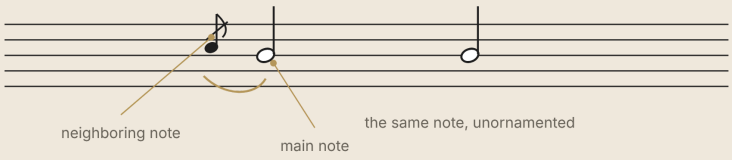
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CHAPTER 18

Appoggiatura

The **appoggiatura** is a musical term for a note used to ornament the main melody. This secondary note may lie below or above the main note, but it must be a neighboring note, usually a second away. It is typically performed with emphasis and then resolves to the main note. There are short appoggiaturas, which resolve quickly, and long ones, which take up a significant part of the main note's value.

APPOGGIATURA: THE NEIGHBORING NOTE THAT ADORNS



The appoggiatura "leans" on the main note from a step above or below.

The appoggiatura: a neighboring note that precedes and enhances the main note.

At first glance, appoggiatura and melisma may seem to be the same thing, but they are not. A melisma may contain several appoggiaturas, but an appoggiatura on its own is not a melisma.

In popular singing, the appoggiatura opens up an enormous range of possibilities for the singer, and melisma itself would not exist without it.

EXERCISE · HOW TO PRACTICE THE APPOGGIATURA

1. First of all, keep your vibrato in good shape: it is the “mother technique” of virtually every ornamentation technique in singing.
2. Sing a section of the original melody.
3. Once the melody is in your head, start looking for spaces where you can add a note; do this one note at a time and memorize everything.
4. Repeat the process in other parts of the song.

This trains your mind to create and memorize interpretations that you build yourself.

Audio for this exercise available for download · see page 154



IV

PART IV

Advanced Techniques

Vocal modes, sound colors, effects, and belting: the technical repertoire of contemporary singing.

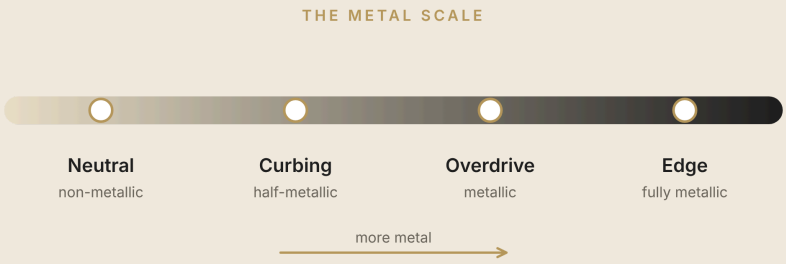
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CHAPTER 19

The Vocal Modes

The voice can be organized into four vocal modes. Together they form a scale of "metal," running from the least metallic timbre of the voice to the most metallic. Think of this scale as the singer's "effects pedalboard": you choose your sound profile according to technical need, personal preference, or the style you are singing. The four vocal modes are **Neutral**, **Curbing**, **Overdrive**, and **Edge**.

Each mode has its own distinct characteristics. Understanding the modes, their qualities, their advantages and their limits, and learning to apply them, is essential for any singer who wants to use their full vocal potential safely and well, to bring color to their singing, and above all to avoid technical problems.



The four vocal modes arranged from the least metallic timbre to the most metallic.

WHERE THE TERM COMES FROM

The organization of the voice into four modes (Neutral, Curbing, Overdrive, and Edge), along with the concepts of “sound colors” and “effects,” was systematized by the Danish singer and researcher Cathrine Sadolin in her Complete Vocal Technique method (SADOLIN, 2012). The mode names are given in English, as in the original method.

THE MODES AT A GLANCE

MODE	METAL	TIMBRE	WHERE IT WORKS BEST
Neutral	Non-metallic	Soft, rounded, velvety	All ranges; in the high range, it takes on a classical, operatic placement
Curbing	Half-metallic	Plaintive, held back	Excellent in the middle range; sounds “flattened” up high
Overdrive	Metallic	Thick, dense, powerful	Low and upper-middle ranges; avoid in the middle range
Edge	Very metallic	Pointed, bright, sharp	High range; not recommended in the middle and low ranges

Neutral

- **Metal:** non-metallic.
- **Timbre:** generally soft, rounded, and velvety. Sound image: the soft, gentle speech of Belo Horizonte, in Brazil.
- **Character:** sad, romantic, introspective, or even stately when more covered; also used when you want the listener to feel the voice close to their ear.
- **Ranges:** every range of the voice. In the high range, however, it will inevitably take on the placement of classical singing.
- **Vowels:** all vowels can be used.
- **Colors:** the sound color can be altered widely: light Neutral, very light Neutral, dark Neutral.

- **Expression:** eyebrows raised; the sound is directed toward the top of the head.
- **Tongue:** the back of the tongue must stay firm and high so the resonance does not fall back; the front, visible part stays flat.
- **Counter-support** set on an /f/: pelvic-floor contraction, engaged lower abdominals, a toned abdomen, lower trapezius, pectorals, a raised sternum, and the upper belt (an expanded neck).

LISTEN

Examples of Neutral: **Emily Kokal** and **Ariana Grande**.

Curbing

- **Metal:** half-metallic.
- **Timbre:** plaintive; or a little more aggressive without going overboard; or brighter and more cheerful than usual, like loose, open, louder speech. Sound image: the plaintive lilt of Brazil's Northeast.
- **Character:** pain, as if you were moaning, lamenting, or complaining. It also suits restrained anger or unrestrained joy.
- **Ranges:** every range of the voice; it is excellent in the middle range but sounds "flattened" up high.
- **Vowels:** ideal for [i]. The other vowels, including [o] and [u], tend toward the vowel sound in *herb*.
- **Colors:** the sound color can be altered widely: light Curbing, square Curbing, dark Curbing, closed Curbing.
- **Expression:** eyebrows relaxed; the sound is aimed straight at the front teeth.
- **Twang:** normal. In dark Curbing, the tip of the tongue may rise, acting as a damper on the harmonics.

– **Counter-support** set on a “ksss”: gluteal contraction, a strong pelvic tuck, a toned abdomen, the lateral obliques, and the middle belt.

LISTEN

Examples of Curbing: **Paula Fernandes** and **Beyoncé**.

Overdrive

– **Metal:** metallic.

– **Timbre:** thick, dense, dark, full, and powerful. Sound image: the dense, emphatic speech of São Paulo.

– **Character:** aggressive, or charged with strong, overt emotions: anger, passion, joy, euphoria.

– **Ranges:** good in the low range and the upper-middle range. Not recommended in the middle range of the voice.

– **Vowels:** Overdrive has no [o], [i], or [u]. The vowels take on a darker, more closed quality, roughly like the “uh” in *hut* and the “eh” in *bed*.

– **Colors:** the sound color can be altered: light Overdrive, dark Overdrive.

– **Expression:** eyebrows raised; the resonance circulates through all the cavities of the head.

– **Twang:** with the tongue resting on the floor of the mouth.

– **Counter-support** set on an /f/ for a more velvety timbre, or on a “ksss” for light Overdrive. For dark Overdrive, it is set on a “chhhh” (as in “shhh”): pelvic floor, glutes, lower back, lower trapezius, and the upper belt (an expanded neck).

LISTEN

Examples of Overdrive: **Taya Smith** and **Beyoncé**.

Edge

- **Metal:** very metallic.
- **Timbre:** extremely metallic, pointed, bright, clear, aggressive, and sharp. Sound image: a strident, piercing sound.
- **Use:** out of necessity or for stylistic reasons, to reach high notes with great volume and a large amount of metal.
- **Ranges:** every range of the voice, although it is not recommended in the middle and low ranges.
- **Vowels:** only twanged vowels can be used: [i] as in *sit*, [æ] as in *and*, [eɪ] as in *stay*, and the vowel in *herb*.
- **Colors:** the sound color cannot be altered much: Edge, square Edge.
- **Expression:** eyebrows drawn together; no covering: the resonance is strictly nasal and/or oral.
- **Twang:** maximum. In some cases, the tip of the tongue may even protrude from the mouth.
- **Counter-support** set on a "ksss."

LISTEN

Examples of Edge: **Taylor Swift** and **Carrie Underwood**.

CAUTION

The more metallic modes (Overdrive and Edge) require command of support, counter-support, and twang. Practice them under guidance, only briefly at first, and stop at the first sign of discomfort. The fundamentals covered in Chapters 2, 9, and 13 are prerequisites.

“

Every person is unique — and will reach unique sound colors and effects.

CHAPTER 20 · SOUND COLORS AND EFFECTS

20

CHAPTER 20

Sound Colors and Effects

Sound colors

Every mode can be altered by shaping the timbre to produce lighter or darker colors, though some modes allow more alteration than others. Colors change through adjustments to the vocal tract (the space just above the vocal folds, extending to the lips and including the nasal passage) and through control of support and counter-support, since the way the chest and abdominal muscles are engaged influences the configuration of the laryngeal muscles (SADOLIN, 2012).

Every singer has a different body, so every person is unique and will arrive at unique sound colors and effects. You can only explore these possibilities for changing the color of your voice once you learn to control each of these physical mechanisms, so that you can engage them while you sing and produce different colors.

Effects

Effects are noises and small ornaments that can be added to the voice or the phrasing to heighten a singer's expression or style. Because every voice is unique, each effect must be built specifically for each singer, taking into account their anatomy, physiology, energy level, and temperament.

BEFORE YOU BEGIN

Before working with effects, you need a firm grasp of the fundamentals of healthy voice use (Parts I and II), as well as a thorough knowledge and command of the modes and their colors.

“

The voice serves the text.

CHAPTER 21 · BELTING

21

CHAPTER 21

Belting

Belting belongs to what is known as *Contemporary Commercial Music* (CCM). The term covers the American techniques of *belting* and *speech-level singing* and applies to the techniques associated with American commercial music, which in Brazil can be heard in gospel music and other segments of pop, as well as in the technique used in Brazilian productions of Broadway musicals. This approach is recent in Brazil, but it has already earned a prominent place in voice teaching because of its strong commercial appeal.

Understanding this style matters because vocal models imported from the United States are often used without rigor or proper training in musicals, stage shows, and Brazilian vocal music, leading to inconsistent timbre, health problems, and a loss of vocal quality.

Registers and belting

To understand belting technique, we must first define registers, an area prone to misunderstanding. In the classic definition by Manuel Garcia (1847), a register is "a series of consecutive and homogeneous tones, from low to high, produced by the same mechanical principle, and whose nature differs essentially from another series of tones, equally consecutive and homogeneous, produced by another mechanical principle."

The term *belting* comes from the verb *to belt out*, meaning to sing in a forced, shouted way. The term is still highly controversial. The voice scientist Ingo Titze speaks of a "production style" rather than a technique. LoVetri (2003, p. 211) states that it is "merely a label given to a certain aspect of chest-register function." Today, belting is recognized as a style or technique of singing in mixed voice (*mix*) with a predominance of the chest sub-register, usually in the upper part of the range. Its timbre is described as having a great deal of brilliance (*ring*), a metallic quality (*brassiness*), *twanginess* (the bright, edgy timbre heard in *country* singers), and loud dynamics.

Origins and history

Belting emerged in the 1930s with singers such as Ethel Merman, Celeste Holm, and Judy Garland, as an alternative to *legit*, the classical style of Broadway singing: less projected and better suited to the microphone, but with a timbre and production very close to those of operatic singing.

In the musical *Oklahoma!*, belting became firmly established as the sound of comic characters and comedy numbers. It has been a source of controversy among voice teachers ever since. Cornelius Reid, one of the leading voice pedagogues of the 1960s, criticized it as "the practice of pop singers singing to the limits of the chest register."

The typical belt of the 1940s can be heard in the voice of Judy Garland: a heavy chest voice that did not go much above C4 and sometimes cracked at A3. It was marked by constant effort, strain, and, often, early retirement.

Today's belt shows greater freedom in sound production, less effort, little vibrato, and a bright (forward) sound, without the earlier sense of precariousness or fragility. The timbre is defined predominantly by the action of the thyroarytenoid (TA) muscle, in a TA-dominant mixed voice (*chest mix*) that can reach G4 in female *high belters*.

The word comes first

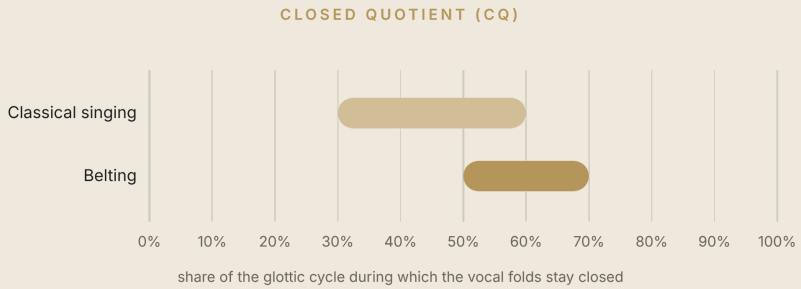
The relationship between singing and speech is intrinsic to Broadway music. Teachers and directors constantly remind performers that this is musical theater: the word, the text, comes first. The voice serves the text. Belting grew out of the need to sing to large audiences without a microphone, with articulation close to that of speech, in a range close to that of the *call*: the loud, almost shouted call, the "hey!"

That is why the music is written in the speaking range (with the exception of *legit*, which follows the rules of classical singing more closely). When the melody moves somewhat away from the speaking range, belting becomes even more important, because it preserves the more natural articulation and delivery of speech. It is always essential that the text be understood the first time (SCHUTTE; MILLER, 1993, p. 142), and the resulting sound must seem natural; a covered or dark voice is ruled out. Belting is the equivalent of "impassioned speech" (SPIVEY, 2008).

Source and filter in belting

Belting has its own characteristics both at the source (phonation) and in the filter (the vocal tract). At the source, the closed quotient of the vocal folds (CQ) in classical singing generally falls between 30% and 60%

(in tenors). In belting, it averages between 50% and 70% for all voice types (WELLS, 2006, p. 69), which reflects greater tension in phonation, higher subglottal pressure, higher sound pressure levels, more tension in the extrinsic muscles, and greater adduction of the vocal folds.

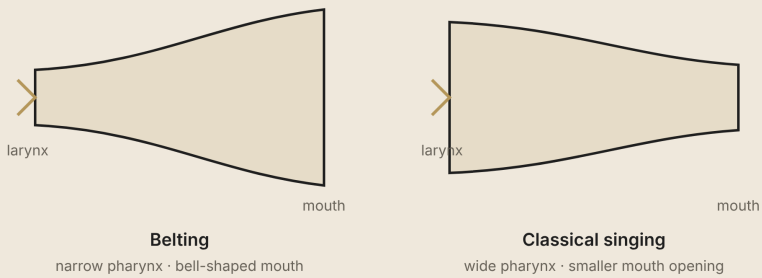


In belting, the vocal folds stay closed for a larger portion of each glottal cycle.

Even so, for all the importance of the phonatory adjustment, the main component of belting is the adjustment of the vocal tract (LOVETRI, 2003), achieved intuitively through a process called *formant tuning*. Belting is also characterized by a slightly raised larynx, a narrower pharynx (POPEIL, 2012), twang (a narrowed aryepiglottic sphincter), and a shortened vocal tract, all of which produce a brighter sound.

The mouth is usually bell-shaped, like a megaphone, with the jaw dropped wide open, unlike classical singing, which is characterized by an inverted megaphone (TITZE; WORLEY; STORY, 2011). Between C4 and E4, classical sopranos reduce their mouth opening; belters do the opposite. Only after the passaggio, once in head voice, do classical singers open the mouth wider.

THE MEGAPHONE AND THE INVERTED MEGAPHONE



Belting and classical singing: the megaphone and the inverted megaphone.

WATCH

Watch **Idina Menzel** perform in belt: you can clearly see the bell-shaped mouth and the wide-open jaw.

For Brazilian singers, it may be easier to learn belting with a foundation in classical singing, since Brazilian urban popular singing is very different from belting. Singing in belt demands great physical and muscular stamina, excellent breath control, and awareness of vocal-tract and phonatory adjustments.

Belting substyles

There are many substyles of belting, and different teachers follow different classifications. The researcher Johan Sundberg proposes the following: *heavy*, *brassy*, *ringy*, nasal, and *speechlike*. Other substyles, tied to particular musicals or repertoires, include Old Disney, New Disney, *rock belt* (as in the musical *Rent*), *soul belt*, *pop belt*, and many more. These substyles still depend heavily on individual teachers, and there is a long way to go before a definitive classification emerges.

Belting in Brazil

From the 2000s onward, another quintessentially American cultural form, the Broadway musical, achieved great success in Brazil as imported shows translated into Portuguese began to be produced. Like gospel music, these shows prize vocal virtuosity and leave little room for colloquial speech. Many of them, such as *Les Misérables* (2001) and *Beauty and the Beast* (2002), are American franchises whose contracts require close fidelity to the originals. On the one hand, they pushed productions to a level of professionalism previously unknown in Brazil; on the other, they imposed a "package" with "all the choreography, arrangements, sets, costumes, and vocal parts. (...) In these shows, nothing or almost nothing seems to be created. Much less improvised" (VENEZIANO, 2008, p. 2-3).

The demand for actor-singers created by these shows spurred interest in studying belting. Many Brazilian teachers specialized in this kind of singing; some imported the procedures of American schools wholesale, even becoming certified to teach commercially franchised methods, while others adapted this knowledge and developed methods of their own (MARIZ, 2013). Because the articulatory and resonance patterns of belting lead to a delivery that tends toward the sound of American English even when sung in Portuguese, the technique needs to be adapted for use in Brazil, and often it is not (ELME; FERNANDES, 2014).

Television also shows a trend toward standardizing the vocal performance of Brazilian popular singers. The success of American talent shows such as *American Idol* and *The Voice* carried over to Brazil through the import of these franchises, renamed *Ídolos* and *The Voice Brasil*, which reproduce not only the format but also a preference for a standardized aesthetic based on American pop singing, especially the gospel-influenced style in which melismas stand out (ELME; FERNANDES, 2014).

The technique of MPB (Brazilian popular music) has nothing to do with belting. A song meant to be sung softly, as if into the listener's ear, as in bossa nova, with the microphone as close as possible, has none of the breath or timbre characteristics associated with opera or musical theater. Like any vocal technique, belting must be adapted to the characteristics of the language, which takes time and study, and when it is used in Brazilian song, it needs to be reassessed. At the same time, there is an urgent need for initiatives that help value the cultural legacy left by Brazil's songwriters and popular singers and bring it back into everyday Brazilian life.

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